



**Petition Definition Document
(DDP)**

***Mortgage Hybrid IRB Models
Implementation Details***

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Version Control

Version	Date	Modification	Author	Sign Off
0.1	14/08/2024	First draft. LGD Only	Retail Methodology	
0.2	13/09/2024	Latest view of PD, EAD, LGD	Retail Methodology	
0.3	24/10/2024	Latest view of PD, EAD, LGD	Retail Methodology	
0.4	31/01/2025	EAD Updates	Retail Methodology	
0.4	14/02/2025	PRA Submission Version	Retail Methodology	

1. Introduction

1.1. Objective

The scope of this document is to provide the IT Capital Reporting team with the details for the Mortgage Hybrid IRB Models implementation into Data Lake. The models are developed using BDR data fields, since it is the historic golden source for IRB related analysis and modelling, and stores the needed historical data. This document specifies risk drivers in each model where they will be required to be implemented into the Data Lake platform. This will require some mapping between Lake and BDR to be considered. Some indication of the mapping may be provided here based on analysis undertaken by the Retail Methodology team but must be reviewed as part of UAT testing to ensure that the models are implemented on the same basis as they were developed. The models will be completely automated in Avalon, and will be updated on a monthly basis to use recent and relevant information. No functionality for overrides is required.

The Mortgage Hybrid IRB Models consist of three models:

- LGD
- EAD
- PD

The tactical code used to run the impact assessment of the new Mortgage Hybrid models is also attached here to provide more details.



Hybrid IRB Data V14.egp

1.2. Common Vocabulary and acronyms

Term	Definition
<i>BDR</i>	<i>Basel Data Repository</i>
<i>PD</i>	<i>Probability of Default</i>
<i>EAD</i>	<i>Exposure at Default</i>
<i>LGD</i>	<i>Loss Given Default</i>

1.3. References

Reference	Document	Location



Reference	Document	Location



2. Business Objectives

2.1. Objectives to be reached with the system

The Mortgage Hybrid IRB models have been developed to satisfy the latest regulatory landscape and provide intuitive and robust risk management to the business with regards to Regulatory and Economic Capital.

3. Models' Descriptions

3.1. PD Functional Requirements

Scope of Mortgage Hybrid PD Model

DDP Ref	DDP Mortgage Hybrid	System	Data Lake
Requirements	The filtering criteria to identify the range of application of the new Hybrid Mortgage PD model is unchanged from the filters that are applied to the mortgage contracts table in the Data Lake to get the applicable contract population for calculating mortgages under the IRB model.		
Additional Information	The mortgage IRB perimeter is as follows: Portfolio: - cntr_portfolio = 'Mortgages' and cntr_subproduct_code in ('402','403','404','405','704','705','707','742','746','999') direct mapping from contract_mortgage_partitions (Retail).partenon_subproduct. cntr_portfolio of 'Mortgages' is set where the contract is sourced from common layer table retail_contracts_enriched.contract_mortgage_partitions, a filter is applied to contract_mortgage_partitions when loading main_contract_partitions where accounting_product_division is not 'STONEHAVEN_MORTGAGE' (also identified with Lender_code 61). Materiality (Balances > £100): - Where the contract is a mortgage i.e. main_contract_partitions.mortgage_cd is 'M' or 'F', then total of ('DRAWN_BALANCE'+ 'ACCRUED_INTEREST'+ 'UNDRAWN_BALANCE') from main_accounting_partitions is greater than 100. - Where the main_contract_partitions.mortgage_cd is 'N' i.e. a new offer the balance criteria is not applied. Asset Class: - asset_class not in (main_contract_partitions.cntr_asset_class = '2' or main_contract_partitions.cntr_asset_class='02') i.e. include where asset class is 1, 01 or null. IRB Account Status and Lender Code: - contract_mortgage_partitions.irb_account_status in ('OP', 'PO') or (contract_mortgage_partitions.irb_account_status = 'PC' and contract_mortgage_partitions.write_off_date is null) - AND NOT (source_system = 'ANMF' AND lender_code in (60,62)) 60 also = migrated UFSS Welcome contracts, 62 = migrated UFSS mortgage plus contracts and unsecured loan parts.		

Sourced PD Model Risk Drivers

DDP Ref	DDP Mortgage Hybrid	System	Data Lake																														
Requirements	Source and ensure inclusion of all PD Model Risk Drivers in retail_capital_enriched.main_mortgage_driver_partitions table.																																
Additional Information	<u>Introduction</u> This section provides a list of the PD model risk drivers that already exist currently in the Data Lake (Either Avalon Capital Engine or Common Layer) and that are required in order to perform the PD Model Calculation. <u>Sourced PD Model Risk Drivers</u> All the fields are listed with their BDR or model development source name in the table below. The equivalent drivers in data lake should be used. This mapping should be checked by the implementation team. Data Lake tables referenced in table below are: Retail_capital_enriched.main_mortgage_driver_partitions (driver) Retail_capital_enriched.mortgage_ahc_output_partitions (ahc) <table><tr><th>Dev/BDR Field</th><th>Existing Lake Field</th><th>Lake Table</th></tr><tr><td>buy_to_let_ind</td><td>buy_to_let_ind</td><td>driver</td></tr><tr><td>period_end_date</td><td>date_processed</td><td>driver</td></tr><tr><td>earliest_maturity</td><td>maturity_date_npl</td><td>ahc</td></tr><tr><td>pra_status</td><td>status_pra</td><td>ahc</td></tr><tr><td>time_on_books_mths_cnt</td><td>time_on_books_mths_cnt</td><td>driver</td></tr><tr><td>io_flag</td><td>io_flag</td><td>driver</td></tr><tr><td>maturity_months</td><td>approaching_maturity_mnth</td><td>driver</td></tr><tr><td>pra_forbearance_flag</td><td>forbearance_flag_pra</td><td>ahc</td></tr><tr><td>app_score</td><td>application_score</td><td>driver</td></tr></table> Table 1: Mortgage Hybrid Sourced PD Risk Drivers Additional Notes:			Dev/BDR Field	Existing Lake Field	Lake Table	buy_to_let_ind	buy_to_let_ind	driver	period_end_date	date_processed	driver	earliest_maturity	maturity_date_npl	ahc	pra_status	status_pra	ahc	time_on_books_mths_cnt	time_on_books_mths_cnt	driver	io_flag	io_flag	driver	maturity_months	approaching_maturity_mnth	driver	pra_forbearance_flag	forbearance_flag_pra	ahc	app_score	application_score	driver
Dev/BDR Field	Existing Lake Field	Lake Table																															
buy_to_let_ind	buy_to_let_ind	driver																															
period_end_date	date_processed	driver																															
earliest_maturity	maturity_date_npl	ahc																															
pra_status	status_pra	ahc																															
time_on_books_mths_cnt	time_on_books_mths_cnt	driver																															
io_flag	io_flag	driver																															
maturity_months	approaching_maturity_mnth	driver																															
pra_forbearance_flag	forbearance_flag_pra	ahc																															
app_score	application_score	driver																															

Derived PD Model Risk Drivers

DDP Ref	DDP Mortgage Hybrid	System	Data Lake						
Requirements	Calculate and add new PD Model Risk Drivers to retail_capital_enriched.main_mortgage_driver_partitions table checking for any clashes with or duplication of existing variables.								
Additional Information	<u>Introduction</u> This section provides a list of the PD model risk drivers that are required to be newly calculated in the Data Lake (either Avalon Capital Engine or Common Layer) and that are required in order to perform the PD Model Calculation. <u>PD Characteristics</u> The newly required fields are listed by name in the table below. <table><tr><th>New Lake Field required as per Dev/BDR Field</th></tr><tr><td>default</td></tr><tr><td>system</td></tr><tr><td>pd_portfolio</td></tr><tr><td>intnl_sys_transfer_ind</td></tr><tr><td>behav_score</td></tr></table> Table 2: Mortgage Hybrid PD Risk Drivers This section provides details of where variables in the table above require some derivation rather than simply be sourced from the appropriate Data Lake table. Additional Notes: default default is derived from the pra_status field: if pra_status in ('PRO','PRR','DEF','PIP','SFS','WRO') then default = 1; else default = 0; system system is derived from the holding_reference field: if substr(holding_reference,1,2) = '09' then system = 'SL'; else if substr(holding_reference,1,2) = 'UF' then system = 'A&L'; else if substr(holding_reference,1,3) = 'NPM' then system = 'NPM'; else if substr(holding_reference,1,4) = 'ANMF' then system = 'San'; else system = 'Err'; <i>Note: A&L should be changed in implementation to AL to avoid the '&' symbol conflict when considering macro variables in SAS for instance.</i> pd_portfolio pd_portfolio is derived from the system, buy_to_let_ind fields:			New Lake Field required as per Dev/BDR Field	default	system	pd_portfolio	intnl_sys_transfer_ind	behav_score
New Lake Field required as per Dev/BDR Field									
default									
system									
pd_portfolio									
intnl_sys_transfer_ind									
behav_score									

```
/* Alliance and Leicester (A&L Closed Book) Residential Portfolio */
if system = 'A&L' and buy_to_let_ind ^= 'Y' then pd_portfolio = 'ARES';
```

```
/* Alliance and Leicester (A&L Closed Book) Buy-to-Let Portfolio */
else if system = 'A&L' and buy_to_let_ind = 'Y' then pd_portfolio = 'ABTL';
```

```
/* Santander UK Buy-to-let (BTL) Portfolio */
else if buy_to_let_ind = 'Y' then pd_portfolio = 'SBTL';
```

```
/* Santander UK Residential Portfolio */
else pd_portfolio = 'SRES';
```

maturity_months

This variable can actually already be sourced as **approaching_maturity_mnths** from the Data Lake tables from retail_capital_enriched.main_mortgage_driver_partitions.

maturity_months represents the time span in complete months from period_date to earliest_maturity. The intck SAS function with a 'month' parameter is used for the estimation. Here are some examples:

- 1)
 - period_end_date: 15JAN2023
 - earliest_maturity: 20MAR2023
 - maturity_months: 2
- 2)
 - period_end_date: 31DEC2023
 - earliest_maturity: 01JAN2024
 - maturity_months: 0
- 3)
 - period_end_date: 31MAR2023
 - earliest_maturity: 01MAY2023
 - maturity_months: 1
- 4)
 - period_end_date: 28FEB2023
 - earliest_maturity: 30APR2024
 - maturity_months: 14

Key points to note:

- Only complete months are counted.
- The day of the month doesn't affect the count unless it completes a full month.
- If the dates are in the same month, the result is 0.
- The function counts month boundaries crossed, not calendar months.

intnl_sys_transfer_ind

intnl_sys_transfer_ind is derived from the following query:

```
SELECT CASE
  WHEN INTL_SYS_TRANS_IND_1 IS NULL
    AND INTL_SYS_TRANS_IND_2 IS NULL THEN INTL_SYS_TRANS_IND_3
  WHEN INTL_SYS_TRANS_IND_1 IS NULL THEN INTL_SYS_TRANS_IND_2
  ELSE INTL_SYS_TRANS_IND_1
END AS intnl_sys_transfer_ind,
contract_id
```

```

FROM (
SELECT CASE
    WHEN FUNDS_DESTINAT_VAR IS NULL THEN ORIG_CORE_TYPE_VAR
    ELSE FUNDS_DESTINAT_VAR
END AS INTL_SYS_TRANS_IND_1,
CASE
    WHEN PURPOSE_CODE_SYS_IN = 'R'
    AND SOLICITOR_CODE IN (999,409,421) THEN 'F'
    ELSE ''
END AS INTL_SYS_TRANS_IND_2,
CASE
    WHEN INTERNAL_TRANSFER = 'Y' THEN 'N'
    ELSE ''
END AS INTL_SYS_TRANS_IND_3,
cont_mor.contract_id
FROM
(SELECT account_number,
contract_id,
CASE
    WHEN FUNDS_DESTINAT IN ('S','M','U') THEN FUNDS_DESTINAT
    ELSE ''
END AS FUNDS_DESTINAT_VAR,
loan_purpose_cd AS PURPOSE_CODE_SYS_IN,
CASE
    WHEN loan_purpose_cd IN ('S','L','E') THEN
        CASE
            WHEN orig_core_type IN ('L','A') THEN 'S'
            ELSE orig_core_type
        END
    ELSE ''
END AS ORIG_CORE_TYPE_VAR,
solicitor_code
FROM retail_contracts_enriched.contract_mortgage_partitions) cont_mor
LEFT JOIN multi_entity_credit_risk_data_analytics_enriched.valuations_mortgages_retail
val_ret
ON val_ret.contract_id = cont_mor.contract_id

```

behav_score

behav_score is derived from the following logic. First the segmentation to estimate the scores is estimated as follows:

```

case
    when arrears_history_cd_irb2 = 0 or ea1c01_max = 'Y'
    then 'UNCLEAN'
    when (e1b07_max in ('2','3','4','5','6')) or
        (e1a03_seg >= 1 and e1a03_seg <= 24) or
        (e1a06_seg >= 1 and e1a06_seg <= 24)
    then 'QUASICLEAN'
    else 'SUPERCLEAN'
end as segment
from retail_capital_enriched.main_mortgage_driver_partitions

```

Then, depending on this segmentation, the score components are calculated as follows:

Segment	Score component	Variable to bin	Binning
UNCLEAN	cscore1	E1A03_MAX	-9999999 -< 0 = 31 0 -< 1 = 32 1 -< 5 = 19 5 -< 13 = 21 13 -< 37 = 23 37 - high = 26 Other = 19
	cscore2	E1A11_MAX	-9999999 -< 0 = 59 0 -< 1 = 74 1 -< 5 = 66 5 -< 7 = 67 7 -< 13 = 68 13 -< 37 = 69 37 - high = 70 Other = 54
	cscore3	E1B10_MAX	-9999999 -< 1 = 29 1 -< 22 = 29 22 -< 324 = 29 324 - high = 29 Other = 29
	cscore4	E1B12_MAX	-9999999 -< 0 = 30 0 -< 1 = 31 1 -< 2 = 26 2 -< 3 = 25 3 - high = 24 Other = 24
	cscore5	E2H09_MAX	-9999999 -< 0 = 33 0 -< 5 = 27 5 - high = 35 Other = 27
	cscore6	MIN_PERIOD_SAVINGS_BAL_AMT	-9999999 -< 50 = 29 50 -< 100 = 32 100 -< 200 = 35 200 -< 550 = 38 550 - high = 42 Other = 29

		cscore7	NDHAC04_MAX	-9999999 -< 0 = 31 0 -< 1 = 25 1 -< 3 = 33 3 -< 7 = 27 7 - high = 21 Other = 21
		cscore8	NDHAC07_MIN	-9999999 -< 0 = 31 0 -< 1 = 28 1 -< 3 = 31 3 -< 7 = 27 7 - high = 23 Other = 23
		cscore9	NDPSD11_MAX_IRB2	-9999999 -< 1 = 31 1 -< 4 = 31 4 -< 7 = 26 7 -< 9 = 20 9 - high = 12 Other = 12
		cscore10	TIME_ON_BOOKS_MTHS_CNT	low -< 25 = 20 25 -< 73 = 25 73 -< 97 = 31 97 - high = 36 Other = 20
		cscore11	WORST_EVER_ARREARS_AMT	low -< 50 = 44 50 -< 600 = 38 600 -< 800 = 32 800 -< 2000 = 25 2000 - high = 19 Other = 19
		cscore12	DAYS_PAST_DUE_MNTHS	low -< 0.3 = 52 0.3 -< 0.5 = 47 0.5 -< 0.7 = 37 0.7 -< 0.9 = 29 0.9 -< 1 = 21 1 -< 1.2 = 15 1.2 -< 1.6 = 6 1.6 -< 1.8 = -3 1.8 -< 2.1 = -13 2.1 - high = -27 Other = -27
		cscore13	ARREARS_HISTORY_CDE_IRB2	0 -< 2 = 86 2 -< 3 = 78 3 -< 4 = 53 4 - high = 65 Other = 45

		cscore14	BALANCE_TO_VALUE	. = 31 low -< 20 = 33 20 -< 60 = 31 60 -< 75 = 29 75 - high = 28 Other = 28
		cscore15	E1B07_MIN	'0', 'U' = 34 '-9', '1', 'N', 'T' = 29 '2', '3', '4', '5', '6' = 24 Other = 24
		cscore16	EA1C01_MAX	'-9', 'N', 'T' = 32 'Y' = -12 Other = -12
	QUASICLEAN	cscore1	E1A02_MAX	-9999999 -< 1 = 55 1 -< 150 = 43 150 - high = 26 Other = 26
		cscore2	E1A10_MIN	-9999999 -< 3 = 56 3 - high = 45 Other = 45
		cscore3	E1B09_MAX	-9999999 -< 4 = 41 4 -< 7 = 50 7 - high = 56 Other = 41
		cscore4	E1B10_MAX	-9999999 -< 1 = 52 1 -< 30 = 52 30 -< 290 = 52 290 -< 440 = 52 440 - high = 52 Other = 52
		cscore5	NDHAC04_MAX	-9999999 -< 1 = 45 1 -< 3 = 54 3 -< 9 = 52 9 - high = 46 Other = 45
		cscore6	NDHAC05_MAX	-9999999 -< 1 = 47 1 -< 4 = 54 4 - high = 39 Other = 39
		cscore7	NDPSD11_MAX_IRB2	-9999999 -< 3 = 55 3 -< 6 = 52 6 -< 8 = 45 8 - high = 33 Other = 33

		cscore8	TIME_ON_BOOKS_MTHS_CNT	low -< 21 = 39 21 -< 48 = 49 48 -< 72 = 55 72 - high = 59 Other = 39
		cscore9	BALANCE_TO_VALUE	. = 50 low -< 66 = 61 66 -< 72 = 51 72 - high = 45 Other = 45
		cscore10	BORROWER_TYPE_ALIGNED	3 -< 4 = 58 1 -< 3 = 50 4 - high = 50 Other = 50
		cscore11	E1B07_MIN	'3', '4', '5', '6', 'N' = 43 '1', '2' = 47 '0', 'U' = 57 Other = 43
		cscore12	E2H08_MAX	'3', '4', '5', '6', 'N' = 48 '1', '2' = 52 '0', 'U' = 57 Other = 48
		cscore13	N/A	0
		cscore14	N/A	0
		cscore15	N/A	0
		cscore16	N/A	0
	SUPERCLEAN	cscore1	NDPSD11_MAX_IRB2	-9999999 -< 0 = 113 0 -< 3 = 81 3 -< 5 = 73 5 -< 7 = 61 7 -< 9 = 51 9 - high = 36 Other = 36
		cscore2	E1A10_MAX	-9999999 -< 0 = 62 0 = 78 1 -< 4 = 61 4 - high = 51 Other = 51

		cscore3	E1B10_MIN_IRB2	-9999999 -< 0 = 64 0 = 72 1 -< 10 = 83 10 -< 32 = 79 32 -< 118 = 72 118 -< 234 = 67 234 -< 362 = 61 362 - high = 53 Other = 53
		cscore4	E2H06_MIN	-9999999 -< 0 = 70.001 0 = 74 1 -< 3 = 80 3 -< 28 = 76 28 -< 160 = 73 160 - high = 70 Other = 70
		cscore5	WORST_EVER_ARREARS_AMT	0 = 79 0.00000000001 -< 300 = 63 300 -< 925 = 51 925 - high = 40 Other = 40
		cscore6	BALANCE_TO_VALUE	. = 72 low -< 38 = 87 38 -< 64 = 81 64 -< 75 = 74 75 -< 85 = 71 85 -< 90 = 65 90 - high = 61 Other = 61
		cscore7	CHG_PERD_END_BAL_L12M	low -< -6000 = 88 -6000 -< -3500 = 82 -3500 -< -1400 = 77 -1400 - high = 71 Other = 71
		cscore8	E1B08_MAX	'1' = 50 '-9' = 58 'N', 'T' = 67 'O', 'U' = 78 Other = 50
		cscore9	SINGLE_JOINT HOLDER_IND	'9', 'S' = 67 'J' = 83 Other = 67
		cscore10	N/A	0
		cscore11	N/A	0

	cscore12	N/A	0
	cscore13	N/A	0
	cscore14	N/A	0
	cscore15	N/A	0
	cscore16	N/A	0
Finally, the behavioural score is estimated as follows: $behav_score = \sum_{i=1}^{16} cscore[i]$			

PD Model Calculation

DDP Ref	DDP Mortgage Hybrid	System	Data Lake
Requirements	Identification of segments and application of parameters/estimates for PD model		

Additional Information	<u>Introduction</u> The calculation of the PD Model pertains to the retail_capital_enriched.main_mortgage_model_partitions table in the Data Lake. A new version of this table is required to execute the Mortgage Hybrid table which will eventually replace the existing model table used to execute Mortgage Variable Scalar. The PD model will be updated on a monthly basis. There are only 2 variables existing in retail_capital_enriched.main_mortgage_model_partitions with the same name as what will be required to be calculated for Mortgage Hybrid version. These 2 are: • pd_segment • pd_reg All newly calculated fields mentioned below will pertain to a Hybrid version of the following Data Lake table: Retail_capital_enriched.main_mortgage_model_partitions (model) <u>PD segmentation</u> Drivers required: • pra_status • time_on_books_mths_cnt • intl_sys_transfer_ind • pd_portfolio • io_flag • maturity_months • pra_forbearance_flag • system • buy_to_let_ind Model Fields calculated: • pd_segment Logic: <pre>if pra_status in ('PRO','PRR','DEF','PIP','SFS','WRO') then pd_segment = '0.DEF'; else if pra_status not in ('PRO','PRR','DEF','PIP','SFS','WRO') and time_on_books_mths_cnt < 3 and (intl_sys_transfer_ind ^= 'U' or substr(pd_portfolio,1,1) = 'A') then pd_segment='1.APP'; else if io_flag in ('I','P') and -3<=maturity_months<=9 then pd_segment='2.IOAM'; else if -3<=maturity_months<=9 then pd_segment='3.CAPAM'; else if pra_forbearance_flag = 1 then pd_segment = '4.FORB'; else if system = 'A&L' and buy_to_let_ind = 'Y' then pd_segment = '5.ABTL'; else if system = 'A&L' and buy_to_let_ind ne 'Y' then pd_segment = '6.ARES'; else if buy_to_let_ind = 'Y' then pd_segment = '7.SBTL'; else pd_segment = '8.SRES';</pre> <u>Final adjusted scores</u> Drivers required:
--------------------------------------	--

- pd_segment
- app_score
- behav_score

Model Fields calculated:

- fin_score

Logic:

/* PD Score alignment defined and applied for key segments */

%let intercept_app = -59.5938;

%let slope_app = 1.1639;

%let intercept_ioam = 383.6887;

%let slope_ioam = 0.2011;

%let intercept_capam = 247.4964;

%let slope_capam = 0.4776;

%let intercept_forb = 283.9785;

%let slope_forb = 0.4370;

if pd_segment = '1.APP' then fin_score = round(&intercept_app. + (app_score *
&slope_app.));

else if pd_segment = '2.IOAM' then fin_score = round(&intercept_ioam. + (behav_score *
&slope_ioam.));

else if pd_segment = '3.CAPAM' then fin_score = round(&intercept_capam. + (behav_score *
&slope_capam.));

else if pd_segment = '4.FORB' then fin_score = round(&intercept_forb. + (behav_score *
&slope_forb.));

else fin_score = behav_score;

PD grades

Drivers required:

- pd_segment

Model Fields required:

- fin_score

Model Fields calculated:

- pd_grade

Logic:

```

if pd_segment in ('0.DEF') then pd_grade = 99;
else if fin_score = . then pd_grade = 1;
else if fin_score <= 515 then pd_grade = 1;
else if fin_score >= 516 and fin_score <= 544 then pd_grade = 2;
else if fin_score >= 545 and fin_score <= 558 then pd_grade = 3;
else if fin_score >= 559 and fin_score <= 570 then pd_grade = 4;
else if fin_score >= 571 and fin_score <= 582 then pd_grade = 5;
else if fin_score >= 583 and fin_score <= 593 then pd_grade = 6;
else if fin_score >= 594 and fin_score <= 604 then pd_grade = 7;
else if fin_score >= 605 and fin_score <= 612 then pd_grade = 8;
else if fin_score >= 613 and fin_score <= 631 then pd_grade = 9;
else if fin_score >= 632 and fin_score <= 647 then pd_grade = 10;
else if fin_score >= 648 and fin_score <= 660 then pd_grade = 11;
else if fin_score >= 661 and fin_score <= 682 then pd_grade = 12;
else if fin_score >= 683 and fin_score <= 698 then pd_grade = 13;
else if fin_score >= 699 and fin_score <= 704 then pd_grade = 14;
else if fin_score >= 705 and fin_score <= 724 then pd_grade = 15;
else if fin_score >= 725 then pd_grade = 16;
else pd_grade = 98;

```

PD Hybrid Estimates

Drivers required:

- pd_segment
- pd_portfolio

Model Fields required:

- pd_grade

Parameter Fields set:

- pd_hybrid
- pd_hybrid_ec

Logic:

pd_segment	pd_portfolio	pd_grade	pd_hybrid	pd_hybrid_e c
'0.DEF'	Any	Any	1	1
Not ('0.DEF')	SRES	1	0.73589	0.73401
		2	0.42581	0.42414

			3	0.20363	0.19912	
			4	0.14719	0.14274	
			5	0.11572	0.11136	
			6	0.09475	0.09035	
			7	0.07554	0.07141	
			8	0.05931	0.05571	
			9	0.04464	0.04149	
			10	0.03325	0.03050	
			11	0.02584	0.02350	
			12	0.01698	0.01515	
			13	0.01304	0.01151	
			14	0.01062	0.00928	
			15	0.00777	0.00658	
			16	0.00547	0.00459	
		SBTL	1	0.75635	0.73401	
			2	0.43802	0.42414	
			3	0.22745	0.20835	
			4	0.16669	0.15043	
			5	0.13134	0.11794	
			6	0.10733	0.09658	
			7	0.08430	0.07786	
			8	0.06698	0.06188	
			9	0.05155	0.04760	
			10	0.03976	0.03642	
			11	0.03192	0.02938	
			12	0.02275	0.02095	
			13	0.01878	0.01731	
			14	0.01645	0.01507	
			15	0.01374	0.01237	
			16	0.01145	0.01038	
		ARES	1	0.74655	0.73401	
			2	0.43051	0.42414	
			3	0.21013	0.19912	
			4	0.15200	0.14274	
			5	0.11922	0.11136	
			6	0.09803	0.09035	
			7	0.07888	0.07141	
			8	0.06281	0.05571	
			9	0.04659	0.04149	
			10	0.03497	0.03050	
			11	0.02726	0.02350	
			12	0.01804	0.01515	
			13	0.01404	0.01151	

			14	0.01154	0.00928
			15	0.00874	0.00658
			16	0.00644	0.00459
		ABTL	1	0.82488	0.73401
			2	0.46465	0.42414
			3	0.27185	0.20835
			4	0.20545	0.15043
			5	0.14726	0.11794
			6	0.13718	0.09658
			7	0.09988	0.07786
			8	0.08193	0.06188
			9	0.05693	0.04760
			10	0.04400	0.03642
			11	0.03454	0.02938
			12	0.02470	0.02095
			13	0.02056	0.01731
			14	0.01836	0.01507
			15	0.01473	0.01237
			16	0.01243	0.01038

Regulatory and Economic PD

Drivers required:

- pd_segment

Parameter Fields required:

- pd_hybrid
- pd_hybrid_ec

Parameter Fields calculated:

- pd_reg
- pd_ec

Logic:

if pd_segment = '0.DEF' then do;

 pd_reg = 1;

 pd_ec = 1;

end;

else do;

 pd_reg = max(0.0003, pd_hybrid);

 pd_ec = pd_hybrid_ec;

end;

3.2. LGD Functional Requirements

Scope of Mortgage Hybrid LGD Model

DDP Ref	DDP Mortgage Hybrid	System	Data Lake
Requirements	The filtering criteria to identify the range of application of the new Hybrid Mortgage LGD model is unchanged from the filters that are applied to the mortgage contracts table in the Data Lake to get the applicable contract population for calculating mortgages under the IRB model.		
Additional Information	The mortgage IRB perimeter is as follows: Portfolio: - cntr_portfolio = 'Mortgages' and cntr_subproduct_code in ('402','403','404','405','704','705','707','742','746','999') direct mapping from contract_mortgage_partitions (Retail).partenon_subproduct. cntr_portfolio of 'Mortgages' is set where the contract is sourced from common layer table retail_contracts_enriched.contract_mortgage_partitions, a filter is applied to contract_mortgage_partitions when loading main_contract_partitions where accounting_product_division is not 'STONEHAVEN_MORTGAGE' (also identified with Lender_code 61). Materiality (Balances > £100): - Where the contract is a mortgage i.e. main_contract_partitions.mortgage_cd is 'M' or 'F', then total of ('DRAWN_BALANCE'+ 'ACCRUED_INTEREST'+ 'UNDRAWN_BALANCE') from main_accounting_partitions is greater than 100. - Where the main_contract_partitions.mortgage_cd is 'N' i.e. a new offer the balance criteria is not applied. Asset Class: - asset_class not in (main_contract_partitions.cntr_asset_class = '2' or main_contract_partitions.cntr_asset_class='02') i.e. include where asset class is 1, 01 or null. IRB Account Status and Lender Code: - contract_mortgage_partitions.irb_account_status in ('OP', 'PO') or (contract_mortgage_partitions.irb_account_status = 'PC' and contract_mortgage_partitions.write_off_date is null) - AND NOT (source_system = 'ANMF' AND lender_code in (60,62)) 60 also = migrated UFSS Welcome contracts, 62 = migrated UFSS mortgage plus contracts and unsecured loan parts.		



Sourced LGD Model Risk Drivers

DDP Ref	DDP Mortgage Hybrid	System	Data Lake																																																								
Requirements	Source and ensure inclusion of all LGD Model Risk Drivers in retail_capital_enriched.main_mortgage_driver_partitions table.																																																										
Additional Information	<u>Introduction</u> This section provides a list of the LGD model risk drivers that already exist currently in the Data Lake (Either Avalon Capital Engine or Common Layer) and that are required in order to perform the LGD Model Calculation.																																																										
	<u>Sourced LGD Model Risk Drivers</u>																																																										
	All the fields are listed with their BDR or model development source name in the table below.																																																										
	The equivalent drivers in data lake should be used. This mapping should be checked by the implementation team.																																																										
	Data Lake tables referenced in table below are:																																																										
	Retail_capital_enriched.main_mortgage_driver_partitions (driver)																																																										
	Retail_capital_enriched.mortgage_ahc_output_partitions (ahc)																																																										
	Retail_contracts_enriched.property_mortgage_partitions (property)																																																										
	Retail_contracts_enriched.mortgage_loan_partitions.pricing (pricing)																																																										
	<table><tr><th>Dev/BDR Field</th><th>Existing Lake Field</th><th>Lake Table</th></tr><tr><td>io_flag</td><td>io_flag</td><td>driver</td></tr><tr><td>maturity_months</td><td>approaching_maturity_mnths</td><td>driver</td></tr><tr><td>on_balance_sheet_exposure</td><td>on_balance_sheet_exposure</td><td>driver</td></tr><tr><td>offer_amt</td><td>offer_amt</td><td>driver</td></tr><tr><td>shared_owner_pcmt</td><td>shared_owner_pcmt</td><td>driver</td></tr><tr><td>pra_status</td><td>status_pra</td><td>ahc</td></tr><tr><td>period_end_date</td><td>date_processed</td><td>driver</td></tr><tr><td>pra_default_entry_date</td><td>def_entry_date_pra</td><td>ahc</td></tr><tr><td>current_avm_confidence</td><td>current_avm_confidence</td><td>driver</td></tr><tr><td>current_valuation_amt_avm</td><td>current_valuation_amt_avm</td><td>driver</td></tr><tr><td>current_valuation_amt</td><td>!</td><td>!</td></tr><tr><td>region_no</td><td>region_no</td><td>driver</td></tr><tr><td>pip_ind</td><td>pip_ind</td><td>driver</td></tr><tr><td>property_typ_cd</td><td>property_typ_cd</td><td>driver</td></tr><tr><td>pra_arr_band</td><td>dpd_n_pra</td><td>driver</td></tr><tr><td>pra_def_reason_cd</td><td>def_reason_cd_pra</td><td>ahc</td></tr><tr><td>property_sale_date</td><td>pip_date_of_sale</td><td>property</td></tr><tr><td>system_cd</td><td>system_cd</td><td>driver</td></tr></table>			Dev/BDR Field	Existing Lake Field	Lake Table	io_flag	io_flag	driver	maturity_months	approaching_maturity_mnths	driver	on_balance_sheet_exposure	on_balance_sheet_exposure	driver	offer_amt	offer_amt	driver	shared_owner_pcmt	shared_owner_pcmt	driver	pra_status	status_pra	ahc	period_end_date	date_processed	driver	pra_default_entry_date	def_entry_date_pra	ahc	current_avm_confidence	current_avm_confidence	driver	current_valuation_amt_avm	current_valuation_amt_avm	driver	current_valuation_amt	!	!	region_no	region_no	driver	pip_ind	pip_ind	driver	property_typ_cd	property_typ_cd	driver	pra_arr_band	dpd_n_pra	driver	pra_def_reason_cd	def_reason_cd_pra	ahc	property_sale_date	pip_date_of_sale	property	system_cd	system_cd
Dev/BDR Field	Existing Lake Field	Lake Table																																																									
io_flag	io_flag	driver																																																									
maturity_months	approaching_maturity_mnths	driver																																																									
on_balance_sheet_exposure	on_balance_sheet_exposure	driver																																																									
offer_amt	offer_amt	driver																																																									
shared_owner_pcmt	shared_owner_pcmt	driver																																																									
pra_status	status_pra	ahc																																																									
period_end_date	date_processed	driver																																																									
pra_default_entry_date	def_entry_date_pra	ahc																																																									
current_avm_confidence	current_avm_confidence	driver																																																									
current_valuation_amt_avm	current_valuation_amt_avm	driver																																																									
current_valuation_amt	!	!																																																									
region_no	region_no	driver																																																									
pip_ind	pip_ind	driver																																																									
property_typ_cd	property_typ_cd	driver																																																									
pra_arr_band	dpd_n_pra	driver																																																									
pra_def_reason_cd	def_reason_cd_pra	ahc																																																									
property_sale_date	pip_date_of_sale	property																																																									
system_cd	system_cd	driver																																																									

sonia	sonia	external
buy_to_let_ind	buy_to_let_ind	driver
Interest_rate	Interest_rate	pricing

Table 3: Mortgage Hybrid Sourced LGD Risk Drivers

Additional Notes:

current_valuation_amt

Although current_valuation_amt is available in the Lake driver table, a version of this variable must be created that is consistent in its indexing approach to that used within the Mortgage Hybrid development. The implemented variable should use the IHS Markit (Halifax) indexing from the original_valuation_amt up until Dec-20 and then the relative (not absolute) indexing should continue thereafter switching to the Hometrack HPI index.

Derived LGD Model Risk Drivers

DDP Ref	DDP Mortgage Hybrid	System	Data Lake
Requirements	Calculate and add new LGD Model Risk Drivers to retail_capital_enriched.main_mortgage_driver_partitions table checking for any clashes with or duplication of existing variables.		

**Additional
Information**
Introduction

This section provides a list of the LGD model risk drivers that are required to be newly calculated in the Data Lake (Either Avalon Capital Engine or Common Layer) and that are required in order to perform the LGD Model Calculation.

LGD Characteristics

The newly required fields are listed by name in the table below.

New Lake Field required as per Dev/BDR Field
default
monthid
default_entry_monthid
def_ovtrrm
Approach_Matur
on_bal_offer_adj
shared_owner_flag
time_since_default
arrear_balance_amt
arrear_balance_amt_3m
Abs_arr_bal_inc
rat_arr_bal_inc
on_balance_sheet_exposure_3m
Abs_on_bal_exp_inc
rat_on_bal_exp_inc
btl
sugg_sale_price

Table 4: Mortgage Hybrid LGD Risk Drivers

This section provides details of where variables in the table above require some derivation rather than simply be sourced from the appropriate Data Lake table.

Additional Notes:
default

default is derived from the pra_status field:

if pra_status in ('PRO','PRR','DEF','PIP','SFS','WRO') then default = 1;
else default = 0;

	monthid monthid is derived from the date_processed field (the monthid corresponding to the execution month): It is defined by a consecutive counter of the number of months between December 1990 and the date_processed date e.g. 384 = Dec-22. default_entry_monthid default_entry_monthid is derived from the def_entry_date_pra field (the default_entry_monthid corresponding to the default entry date month under the PRA default definition): It is defined by a consecutive counter of the number of months between December 1990 and the def_entry_date_pra date e.g. 384 = Dec-22. def_ovtrm def_ovtrm is derived from the pra_def_reason_cd field: if not missing(pra_def_reason_cd) then do; if substr(pra_def_reason_cd,2,1) = "1" then def_ovtrm = "1"; else def_ovtrm = "0"; end; else do; def_ovtrm="M"; end; Please note that when using the field from the Data Lake this is an 11 character string as opposed to the development data which used a 10 character string. This means the above logic should be altered to look at position 3 instead of position 2: if not missing(def_reason_cd_pra) then do; if substr(def_reason_cd_pra,3,1) = "1" then def_ovtrm = "1"; else def_ovtrm = "0"; end; else do; def_ovtrm="M"; end; Approach_Matur Approach_Matur is derived from the maturity_months field: if -3 <= maturity_months <= 9 then Approach_Matur = 1; else Approach_Matur = 0; on_bal_offer_adj on_bal_offer_adj is derived from the default, on_balance_sheet_exposure, and offer_amt fields: if default = 0 then on_bal_offer_adj = max(sum(on_balance_sheet_exposure, offer_amt), 0); else on_bal_offer_adj = max(on_balance_sheet_exposure, 0); shared_owner_flag shared_owner_flag is derived from the shared_owner_pcmt field: if shared_owner_pcmt in (., 0, 100) then shared_owner_flag = 0; else shared_owner_flag = 1; <i>The . in SAS means missing</i> time_since_default time_since_default is derived from the default, monthid, and default_entry_monthid fields: if default = 1 then do;
--	---

```

time_since_default = monthid - default_entry_monthid;
end;
else do;
    time_since_default = .;
end;
A default value other than a missing value may be required for Lake implementation

arrears_balance_amt
arrears_balance_amt is derived from the source_system, arrears_balance_amt_calc_dpd,
consolidated_overpayments_balanc:
case
    when ctr_part.source_system in ('PARTENON_RETAIL','MACDB') then
        coalesce(ctr_mnth_part.arrears_balance_amt_calc_dpd,0)
    when ctr_part.source_system in ('ANMF','UFSS') then
        coalesce(ctr_mnth_part.arrears_balance_amt_calc_dpd,0) +
        coalesce(ctr_part.consolidated_overpayments_balanc,0)
    else 0
end as ARREARS_BALANCE_AMT format=14.2,

from retail_contracts_enriched.contract_mortgage_partitions ctr_part
left outer join retail_contracts_enriched.contract_mortgage_month_partitions ctr_mnth_part
on ctr_part.contract_id = ctr_mnth_part.contract_id

arrears_balance_amt_3m
arrears_balance_amt_3m should be set as the value of the arrears_balance_amt
arrears_balance_amt field from 3 months prior to the current execution month e.g if
executing/running calculation in Jun-24, the value of arrears_balance_amt_3m will be the
value of the arrears_balance_amt field from Mar-24.

Abs_arr_bal_inc
Abs_arr_bal_inc is derived from the arrears_balance_amt and arrears_balance_amt_3m
fields:
Abs_arr_bal_inc = arrears_balance_amt - arrears_balance_amt_3m;

rat_arr_bal_inc
rat_arr_bal_inc is derived from the Abs_arr_bal_inc and arrears_balance_amt_3m
fields and is defined as the absolute ratio of the arrears balance difference and the arrears
balance from 3 months prior:
rat_arr_bal_inc = ABS(Abs_arr_bal_inc/arrears_balance_amt_3m);
if Abs_arr_bal_inc = 0 then rat_arr_bal_inc = 0;
else if arrears_balance_amt_3m = 0 then rat_arr_bal_inc = 1;
else if Abs_arr_bal_inc < 0 then rat_arr_bal_inc = -1*rat_arr_bal_inc;
rat_arr_bal_inc = round(rat_arr_bal_inc,0.01);

To understand the above SAS code better, there is a field construction logic:


- Step 1 : Take difference between the value at observation period and lagged 3m value
- Step 2 : Convert it to relative change taking ratio with respect to the lagged 3m value
- Step 3 : If Change = 0 then Ratio = 0
- Step 4 : If denominator = 0 then Ratio = 1
- Step 5 : Take absolute value of the Ratio and assign the sign of change based on the sign of absolute change

```

on_balance_sheet_exposure_3m

on_balance_sheet_exposure_3m should be set as the value of the on_balance_sheet_exposure field from 3 months prior to the current execution month e.g if executing/running calculation in Jun-24, the value of on_balance_sheet_exposure_3m will be the value of the on_balance_sheet_exposure field from Mar-24.

Abs_on_bal_exp_inc

Abs_on_bal_exp_inc is derived from the on_balance_sheet_exposure and on_balance_sheet_exposure_3m fields:

$$\text{Abs_on_bal_exp_inc} = \text{on_balance_sheet_exposure} - \text{on_balance_sheet_exposure_3m};$$

rat_on_bal_exp_inc

rat_on_bal_exp_inc is derived from the Abs_on_bal_exp_inc field and on_balance_sheet_exposure_3m fields and is defined as the absolute ratio of the on balance difference and the on balance from 3 months prior:

$$\text{rat_on_bal_exp_inc} = \text{ABS}(\text{Abs_on_bal_exp_inc}/\text{on_balance_sheet_exposure_3m});$$

if Abs_on_bal_exp_inc = 0 then rat_on_bal_exp_inc = 0;

else if on_balance_sheet_exposure_3m = 0 then rat_on_bal_exp_inc = 1;

else if Abs_on_bal_exp_inc < 0 then rat_on_bal_exp_inc = -1*rat_on_bal_exp_inc;

$$\text{rat_on_bal_exp_inc} = \text{round}(\text{rat_on_bal_exp_inc}, 0.01);$$

To understand the above SAS code better, there is a field construction logic:

1. Step 1 : Take difference between the value at observation period and lagged 3m value
2. Step 2 : Convert it to relative change taking ratio with respect to the lagged 3m value
3. Step 3 : If Change = 0 then Ratio = 0
4. Step 4 : If denominator = 0 then Ratio = 1
5. Step 5 : Take absolute value of the Ratio and assign the sign of change based on the sign of absolute change

btl

btl is derived from the buy_to_let_ind field:

if buy_to_let_ind = 'Y' then btl = 'Y'; else btl = 'N';

max_int_rate

The max_int_rate must be obtained by transforming interest_rate from the pricing table. It is defined as the minimum interest rate applied to any portion of a loan for a given period end date. In SAS, this is as follows:

$$\text{max}(\text{case when period_end_balance_amt} > 0 \text{ and interest_rate} > 0 \text{ then interest_rate else } 0 \text{ end}) \text{ as max_int_rate}$$

group by contract_id, period_date

min_int_rate

The min_int_rate must be obtained by transforming interest_rate from the pricing table. It is defined as the minimum interest rate applied to any portion of a loan for a given period end date. In SAS, this is as follows:

$$\text{min}(\text{case when period_end_balance_amt} > 0 \text{ and interest_rate} > 0 \text{ then interest_rate else } 99 \text{ end}) \text{ as min_int_rate}$$

group by contract_id, period_date

sugg_sale_price



	This is the expert valuation for sale of a property in possession and data is currently taken tactically from PIPPROD. Sourcing this variable through into the Data Lake will need to be considered in implementation.
--	---

LGD Model Calculation

DDP Ref	DDP Mortgage Hybrid	System	Data Lake
Requirements	Identification of segments and application of parameters/estimates for LGD model		

Additional Information	<u>Introduction</u> The calculation of the LGD Model pertains to the retail_capital_enriched.main_mortgage_model_partitions table in the Data Lake. A new version of this table is required to execute the Mortgage Hybrid table which will eventually replace the existing model table used to execute Mortgage Variable Scalar. There are only 8 variables existing in retail_capital_enriched.main_mortgage_model_partitions with the same name as what will be required to be calculated for Mortgage Hybrid version. These 8 are: • lgd_cost_of_coll_dt • lgd_dt • lgd_recovery_amt_disc_dt • lgd_recovery_ead2 (derived in EAD model calculation) • lgd_recovery_valuation • lgd_severity_dt • lgd_severity_value_dt • vat_rate Only vat_rate would be consistent in terms of value and therefore not require updating. All newly calculated fields mentioned below will pertain to a Hybrid version of the following Data Lake table: Retail_capital_enriched.main_mortgage_model_partitions (model) <u>Collateral Value</u> Drivers required: • current_avm_confidence • current_valuation_amt_avm • current_valuation_amt • pip_ind • sugg_sale_price Model Fields calculated: • lgd_recovery_valuation • lgd_avm_hpi_flag Logic: <pre> if default = 1 and pip_ind = "Y" and sugg_sale_price ne . then do; lgd_recovery_valuation = sugg_sale_price; lgd_avm_hpi_flag = 'AVM'; end; else if current_avm_confidence in (., 0) or current_valuation_amt_avm in (., 0) then do; lgd_recovery_valuation = current_valuation_amt; lgd_avm_hpi_flag = 'HPI'; </pre>
------------------------	---

```

end;
else do;
    lgd_recovery_valuation = current_valuation_amt_avm;
    lgd_avm_hpi_flag = 'AVM';
end;

```

Haircut

Drivers required:

- default
- region_no
- current_avm_confidence

Model Fields calculated:

- lgd_recovery_segment

Parameter Fields set:

- lgd_haircut_dt
- lgd_haircut_pmdt
- lgd_haircut_beel
- lgd_haircut_lra

Note parameter values have not been finalised yet

Parameter values should take a decimal format of 10.6

Logic:

Not-in-default (default=0) lgd_recovery_segment

Segment	Region	Current AVM Confidence	Haircut dt	Haircut pmdt	Haircut beel	Haircut lra
1	Not (3,7,8)	>=6.5	0.75902	0.76131	0.80677	0.80636
2		5.5<= & <6.5	0.74188	0.74402	0.79034	0.79017
3		Missing or <5.5	0.70543	0.70759	0.76232	0.76178
4	3,8	>=5.5	0.77560	0.77784	0.85917	0.85878
5		Missing or <5.5	0.73431	0.73655	0.80678	0.80568
6	7	Any	0.71182	0.74321	0.71202	0.71182

In-default (default=1) lgd_recovery_segment

Segment	Region	Current AVM Confidence	Haircut dt	Haircut pmdt	Haircut beel	Haircut lra
7	Not (3,7,8)	>=6.5	0.75647	0.75858	0.81827	0.81807
8		5.5<= & <6.5	0.73438	0.73642	0.79647	0.79641

9		Missing or <5.5	0.70181	0.7039	0.76233	0.76209
10	3,8	>=5.5	0.77181	0.77408	0.87278	0.87258
11		Missing or <5.5	0.72797	0.73026	0.81675	0.81613
12	7	Any	0.70241	0.75366	0.70247	0.70241
13 – where pip_ind = "Y" and sugg_sale _price ne .	Any	Any	0.972164	0.972164	1.02815	1.02815

Indirect Costs

Parameter Fields set:

- lgd_cost_collections_mth_[dt/pmdt/beel/lra]

Logic:

lgd_cost_collections_mth_dt = 23.73;
 lgd_cost_collections_mth_pmdt = 23.73;
 lgd_cost_collections_mth_beel = 23.73;
 lgd_cost_collections_mth_lra = 23.73;

Litigation Costs

Parameter Fields set:

- lgd_cost_lit_[dt/pmdt/beel/lra]

Logic:

lgd_cost_lit_beel = 2166.94;
 lgd_cost_lit_dt = 2166.94;
 lgd_cost_lit_pmdt = 2166.94;
 lgd_cost_lit_lra = 2166.94;

Cure Litigation Fees

Parameter Fields set:

- lgd_cure_cost_lit_[dt/pmdt/beel/lra]

Logic:

lgd_cure_cost_lit_beel = 0;
 lgd_cure_cost_lit_dt = 0;
 lgd_cure_cost_lit_pmdt = 0;

	<pre> lgd_cure_cost_lit_lra = 0; <u>Possessions Costs</u> Drivers required: - region_no Parameter Fields set: - lgd_cost_poss_[dt/pmdt/beel/lra] Logic: /*Scotland or Missing*/ if region_no in (7,.) then do; lgd_cost_poss_beel = 581.10; lgd_cost_poss_dt = 581.10; lgd_cost_poss_pmdt = 581.10; lgd_cost_poss_lra = 581.10; end; /*Other*/ else do; lgd_cost_poss_beel = 501.82; lgd_cost_poss_dt = 501.82; lgd_cost_poss_pmdt = 501.82; lgd_cost_poss_lra = 501.82; end; <u>Sales Costs</u> Drivers required: - shared_owner_flag - region_no - property_typ_cd Parameter Fields set: - lgd_cost_sale_[dt/pmdt/beel/lra] Logic: /*Greater London, SE or Missing*/ if region_no in (3,8,.) then do; lgd_cost_sale_beel = 8091.91; lgd_cost_sale_pmdt = 8091.91; lgd_cost_sale_dt = 8373.74; </pre>
--	---

	<pre> lgd_cost_sale_lra = 8373.74; end; /*Other*/ else do; lgd_cost_sale_beel = 3672.50; lgd_cost_sale_pmdt = 3672.50; lgd_cost_sale_dt = 3715.96; lgd_cost_sale_lra = 3715.96; end; Shared ownership uplift: if shared_owner_flag=1 then do; lgd_cost_sale_beel = lgd_cost_sale_beel*2.24; lgd_cost_sale_pmdt = lgd_cost_sale_pmdt*2.24; lgd_cost_sale_dt = lgd_cost_sale_dt*2.24; lgd_cost_sale_lra = lgd_cost_sale_lra*2.24; end; </pre> <u>Selling Commission</u> Parameter Fields set: • lgd_cost_sale_comm_min • lgd_cost_sale_comm_fraction Logic: <pre> lgd_cost_sale_comm_min = 750; lgd_cost_sale_comm_fraction = 0.01; </pre> <u>HPI/AVM Factors</u> Drivers required: • region_no Model Fields required: • lgd_avm_hpi_flag Model Fields calculated: • lgd_hpi_segment Parameter Fields set: • lgd_lgd_hpi_stress_dt
--	--

Note these parameter values are dynamic and as such require their own calculation and own implementation documentation. The parameter values update on a quarterly basis.



HPI AVM DT Factors
v0.1.docx

Logic:

lgd_hpi_segment

Segment	AVM/HPI Flag	Region	Dynamic Parameter Value lgd_lgd_hpi_stress_dt
H1	HPI	1	
H2		2	
H3		3	
H4		4	
H5		5	
H6		6	
H7		7	
H8		8	
H9		9	
H10		10	
H11		11	
H12		12	
H13		13	
A1	AVM	1	
A2		2	
A3		3	
A4		4	
A5		5	
A6		6	
A7		7	
A8		8	
A9		9	
A10		10	
A11		11	
A12		12	
A13		13	

BEEL/DT Recovery Valuation

Drivers required:

- shared_owner_flag
- shared_owner_pcmt

Model Fields required:

- lgd_recovery_valuation
- lgd_lgd_hpi_stress_dt

Model Fields calculated:

- lgd_recovery_valuation_[dt/pmdt/beel/lra]

Logic:

if shared_owner_flag = 1 then do;

lgd_recovery_valuation_beel = lgd_recovery_valuation * shared_owner_pcmt / 100;

lgd_recovery_valuation_dt = lgd_recovery_valuation * (shared_owner_pcmt / 100)
* lgd_lgd_hpi_stress_dt;

end;

else do;

lgd_recovery_valuation_beel = lgd_recovery_valuation;

lgd_recovery_valuation_dt = lgd_recovery_valuation * lgd_lgd_hpi_stress_dt;

end;

lgd_recovery_valuation_pmdt = lgd_recovery_valuation_dt;

lgd_recovery_valuation_lra = lgd_recovery_valuation_beel;

LTV/DLTV

Drivers required:

- on_bal_offer_adj

Model Fields required:

- lgd_recovery_valuation
- lgd_recovery_valuation_beel
- lgd_recovery_valuation_dt

Model Fields calculated:

- ltv2
- dlrv

Logic:

if lgd_recovery_valuation > 0 then ltv2 = 100 * round(on_bal_offer_adj /
lgd_recovery_valuation_beel, 0.01);

else ltv2 = -999;

if lgd_recovery_valuation > 0 then dltv = 100 * round(on_bal_offer_adj / lgd_recovery_valuation_dt, 0.01);

else dltv = -999;

PPGD

Drivers required:

- default
- dltv
- ltv2
- time_since_default

Model Fields calculated:

- lgd_ppgd_segment
- lgd_ppgd_segment_beel

Parameter Fields set:

- lgd_ppgd_[dt/pmdt/beel/lra]

Logic:

Not-in-default (default=0) lgd_ppgd_segment

Segment	DLTV	PPGD dt	PPGD pmdt	PPGD beel	PPGD lra
1	<=30	0.02677	0.02652	0.02427	0.02439
2	30<&<=60	0.07985	0.07943	0.0718	0.07218
3	60<&<=80	0.20034	0.19797	0.14876	0.14935
4	80<&<=90	0.31567	0.3133	0.21183	0.21197
5	>90 or Missing	0.36816	0.36531	0.29727	0.29748

In-default (default=1) lgd_ppgd_segment

Segment	PIP Ind	DLTV	Time Since Default	PRA Arrears Band	PPGD dt	PPGD pmdt	PPGD beel	PPGD lra
6	N	<=30	Missing or <=12	Any	0.03473	0.03468	0.03468	0.03473
7			>12	Any	0.03849	0.03845	0.03845	0.03849
8		30<&<=60	Missing or <=12	Any	0.08478	0.08457	0.08457	0.08478
9			>12	DPD000 DPD030 DPD060 DPD090	0.05913	0.05898	0.04578	0.04579

	10		60<&=<=80	>12	Not (DPD000 DPD030 DPD060 DPD090)	0.15465	0.1537	0.12423	0.12438
	11			Missing or <=12	Any	0.20030	0.19853	0.15275	0.15316
	12			>12	DPD000 DPD030 DPD060 DPD090	0.11068	0.11018	0.06866	0.06870
	13			>12	Not (DPD000 DPD030 DPD060 DPD090)	0.26565	0.26342	0.17712	0.17738
	14		80<&=<=90	Missing or <=12	Any	0.31570	0.31417	0.22208	0.22220
	15			>12	DPD000 DPD030 DPD060 DPD090	0.15854	0.15791	0.09039	0.09043
	16			>12	Not (DPD000 DPD030 DPD060 DPD090)	0.35852	0.35599	0.24823	0.24853
	17		>90 or Missing	Missing or <=12	Any	0.37017	0.36855	0.36738	0.36758
	18			>12	DPD000 DPD030 DPD060 DPD090	0.21189	0.20974	0.15065	0.15073
	19			>12	Not (DPD000 DPD030 DPD060 DPD090)	0.44400	0.44178	0.40300	0.40338
	20	Y	Any	Any	Any	1.00000	1.00000	1.00000	1.00000

Logic to derive lgd_ppgd_segment_beel is the same as the above tables except ltv2 is used instead of dltv

TTP

Drivers required:

- default
- on_bal_offer_adj
- dltv

- ltv2
- pip_ind
- arrears_balance_amt
- rat_on_bal_exp_inc
- rat_arr_bal_inc

Model Fields calculated:

- lgd_ttp_segment
- lgd_ttp_segment_beel

Parameter Fields set:

- lgd_ttp_dt
- lgd_ttp_pmdt
- lgd_ttp_beel
- lgd_ttp_lra

Logic:

lgd_ttp_segment

Not-in-default (default=0)

Segment	DLTV	on_bal_off er _adj	TTP dt	TTP pmdt	TTP beel	TTP lra
1	<= 70 or missing	<40000 or missing	0.00	0.00	0.00	0.00
2	<= 70	>= 40000	49.99698	49.99698	40.29578	40.29578
3	>70	Any	37.23694	37.23694	35.5945	35.5945

Logic to derive lgd_ttp_segment_beel non-default is the same as the above table except ltv2 is used instead of dlvt.

In-default (default=1)

Seg	Pip_ind	rat_on_ bal_ exp_inc	arrears _balance _amt	rat_arr _bal_inc	TTP dt	TTP pmdt	TTP beel	TTP lra
0	Y	Any	Any	Any	30.22187	30.22187	29.2422	29.2422
4	N	<0.001	< 2660	<0.140	55.43744	55.43744	51.27534	51.27534

	5				≥ 0.140	52.42109	52.42109	46.56509	46.56509
	6			≥ 2660	Any	44.62993	44.62993	41.9723	41.9723
	7			< 2660	Any	37.66295	37.66295	34.63024	34.63024
	8		≥ 0.001	≥ 2660	Any	22.15094	21.67748	22.15094	22.15094

The **lgd_ttp_segment_beel** in-default is the same as the above table.
lgd_ttp_segment_beel = lgd_ttp_segment;

TPS

Parameter Fields set:

- **lgd_tps_[dt/pmdt/beel/lra]**

Logic:

lgd_tps_dt = 4.6038;
 lgd_tps_pmdt = 4.6038;
 lgd_tps_beel = 4.2684;
 lgd_tps_lra = 4.2684;

TTS

Parameter Fields required:

- **lgd_ttp_[dt/pmdt/beel/lra]**
- **lgd_tps_[dt/pmdt/beel/lra]**

Parameter Fields set:

- **lgd_tts_[dt/pmdt/beel/lra]**

Logic – note “ceil” here is used as if it is the SAS CEIL function, i.e. to round up a numeric value to the nearest integer:

lgd_tts_dt = ceil(lgd_ttp_dt + lgd_tps_dt);
 lgd_tts_pmdt = ceil(lgd_ttp_pmdt + lgd_tps_pmdt);
 lgd_tts_beel = ceil(lgd_ttp_beel + lgd_tps_beel);
 lgd_tts_lra = ceil(lgd_ttp_lra + lgd_tps_lra);

Cash recovery

Drivers required:

- default
- io_flag

Parameter Fields set:

- lgd_cash_recover_[dt/pmdt/beel/lra]

Logic:

```

if default = 0 then do;
    if io_flag = 'R' then do;
        lgd_cash_recover_dt = 0.0590;
        lgd_cash_recover_pmdt = 0.0590;
        lgd_cash_recover_beel = 0.0590;
        lgd_cash_recover_lra = 0.0590;
    end;
    else do;
        lgd_cash_recover_dt = 0.0332;
        lgd_cash_recover_pmdt = 0.0332;
        lgd_cash_recover_beel = 0.0332;
        lgd_cash_recover_lra = 0.0332;
    end;
end;
else do;
    if io_flag = 'R' then do;
        lgd_cash_recover_dt = 0.0883;
        lgd_cash_recover_pmdt = 0.0883;
        lgd_cash_recover_beel = 0.0905;
        lgd_cash_recover_lra = 0.0905;
    end;
    else do;
        lgd_cash_recover_dt = 0.0485;
        lgd_cash_recover_pmdt = 0.0485;
        lgd_cash_recover_beel = 0.0485;
        lgd_cash_recover_lra = 0.0485;
    end;
end;
end;

```

Discounting rate

Parameter Field set:

- sonia

For a given month, the maximum is taken of either the month-end value or the mean value across the month.

Note the SONIA rate parameter value is dynamic and taken from the Bank of England Website. This will require to be fed into the Data Lake for implementation.

cust_int_rate

Drivers required:

- max_int_rate
- min_int_rate

Parameter Field required:

- sonia

Model Field calculated:

- cust_int_rate

Logic:

if max_int_rate ne 0 then cust_int_rate = max_int_rate/100;
 else if max_int_rate = 0 and min_int_rate not in (99,..) then cust_int_rate = min_int_rate/100;
 else cust_int_rate = sum(sonia,0.03);

Discounting rate - beel

Parameter Field required:

- sonia

Parameter Field calculated:

- lgd_disc_rate_annual_beel

Logic:

lgd_disc_rate_annual_beel = sum(sonia, 0.05);

PRA discount rate – floor 9%

Parameter Field required:

- lgd_disc_rate_annual_beel

Parameter Field calculated:

- lgd_disc_rate_annual_dt

Logic:

lgd_disc_rate_annual_dt = max(0.09, lgd_disc_rate_annual_beel);

Monthly Discount Factor

Model Field required:

	• cust_int_rate Parameter Fields required: • lgd_disc_rate_annual_beel • lgd_disc_rate_annual_dt Model Field calculated: • lgd_disc_rate_cust Parameter Fields calculated: • lgd_disc_rate_monthly_[dt/beel/pmdt/lra] Logic: $\text{lgd_disc_rate_monthly_beel} = (1 + \text{lgd_disc_rate_annual_beel})^{**}(1 / 12);$ $\text{lgd_disc_rate_monthly_dt} = (1 + \text{lgd_disc_rate_annual_dt})^{**}(1 / 12);$ $\text{lgd_disc_rate_monthly_pmdt} = \text{lgd_disc_rate_monthly_dt};$ $\text{lgd_disc_rate_monthly_lra} = \text{lgd_disc_rate_monthly_beel};$ $\text{lgd_disc_rate_cust} = (1 + \text{cust_int_rate})^{**}(1/12);$ <u>VAT rate</u> Parameter Fields set: • vat_rate Note there have been no recent VAT rate changes but this parameter value is dynamic and the current VAT rate should be taken. <u>Loss Given Cure</u> Drivers required: • default • mortgage_cd • approach_matur • pra_status Model Fields required: • on_bal_offer_adj • def_ovtrm Model Fields calculated: • lgd_cure_severity_segment Parameters Fields set: • lgd_lgc_[dt/pmdt/beel/lra] Logic: lgd_cure_severity_segment Not-in-default (default=0)
--	---

Segme nt	Condition	on_bal_ offer_a dj	lgd_lgc_ dt	lgd_lgc_ pmdt	lgd_lgc_ beel	lgd_lgc_ lra
1_1	(mortgage_cd ne 'N' and missing(approach_Matur)) or approach_Matur = 1	Any	0.11628	0.1141	0.09403	0.09506
1_2	Not [(mortgage_cd ne 'N' and missing(approach_Matur)) or approach_Matur = 1]	< 30000	0.08853	0.08751	0.07877	0.07912
1_3		>= 30000 and < 75000	0.02197	0.02165	0.01612	0.01622
1_4		>= 75000	0.00913	0.00897	0.00698	0.00702

In-default (default=1)

Segme nt	def_ ovtrm	on_bal_ offer_adj	pra_stat us	lgd_lgc_ dt	lgd_lgc_ pmdt	lgd_lgc_ beel	lgd_lgc_ lra
2_1	M	Any	Any	0.19323	0.18897	0.12741	0.12905
	1	>= 100					
2_2	Not M nor 1	Missing or < 30000	Any	0.08121	0.08028	0.06733	0.06759
2_3		>= 30000 and < 75000	Not ('PRO' or 'PRR')	0.02767	0.02735	0.02597	0.02607
2_4		>= 30000	'PRO' or 'PRR'	0.0093	0.00914	0.00643	0.00646
2_5		>=75000	Not ('PRO' or 'PRR')	0.01035	0.01013	0.00957	0.00964
2_6	Not M	<= 100	Any	1.0000	1.0000	1.0000	1.0000

MOC values

Parameters Fields set:

- al_moc_[dt/pmdt/beel/lra]
- a9_moc_[dt/pmdt/beel/lra]

Logic:

al_moc_dt = 1.076295;

al_moc_pmdt = 1.070417

```

al_moc_beel = 1.070417;
al_moc_lra = 1.076295;

if default=0 then do;
    a9_moc_dt=1.013344;
    a9_moc_pmdt= 1.007382;
    a9_moc_lra=1.009957;
    a9_moc_beel= 1.005924 ;
end;else do;
    a9_moc_dt=1.010494;
    a9_moc_pmdt= 1.004848 ;
    a9_moc_lra=1.004688;
    a9_moc_beel= 1.000912;
end;

```

Accumulated Cost of Collections

Drivers required:

- default

Model Fields required:

- lgd_tts_[dt/pmdt/beel/lra]

Parameters Fields required:

- lgd_cost_collections_mth_[dt/pmdt/beel/lra]
- lgd_disc_rate_cust

Model Fields calculated:

- lgd_cost_of_coll_[dt/pmdt/beel/lra]

Function (macro in SAS) required: This function estimates the finite sum of a geometric series and serves as the discounting factor used to determine the present value of a series of regular payments. It adjusts for the interest rate and the number of periods, allowing for the accurate valuation of future cash flows in today's terms.

%macro sum_geo_series(rate=,time=);

((&rate.)**(&time.)-1)/((&rate.)**(&time.)*(&rate.-1))

%mend sum_geo_series;

Logic:

if default = 1 then lgd_cost_of_coll_&type. = round(lgd_cost_collections_mth_&type. *
%sum_geo_series(rate=lgd_disc_rate_cust,time=lgd_tts_&type.), **0.01**);

else if default = 0 then lgd_cost_of_coll_&type. = round(lgd_cost_collections_mth_&type. *
(4 + %sum_geo_series(rate=lgd_disc_rate_cust,time=lgd_tts_&type.)), **0.01**);

&type. is replaced by [dt/pmdt/beel/lra] respectively.

Property sale – agents fees and commissions

Model Fields required:

- lgd_recovery_valuation_[dt/pmdt/beel/lra]

Parameters Fields required:

- lgd_cost_sale_comm_min
- lgd_cost_sale_comm_fraction
- lgd_haircut_[dt/pmdt/beel/lra]
- lgd_shared_owner_adj_[dt/pmdt/beel/lra]
- vat_rate

Model Fields calculated:

- lgd_cost_sale_variable_[dt/pmdt/beel/lra]

Logic:

if lgd_recovery_valuation_&type. = . then lgd_cost_sale_variable_&type. = .;
 else lgd_cost_sale_variable_&type. = round(max(lgd_cost_sale_comm_min,
 lgd_cost_sale_comm_fraction * lgd_haircut_&type. * lgd_shared_owner_adj_&type.*
 lgd_recovery_valuation_&type.) * (1 + vat_rate), 0.01);
 . in SAS represents a missing value. &type. is replaced by [dt/beel/pmdt/lra] respectively.

Cash Recovery Amount

Model Fields required:

- lgd_recovery_ead2 (derived in EAD model calculation) for dt, pmdt, lra
- lgd_recovery_ead1 (derived in EAD model calculation) for beel

Parameters Fields required:

- lgd_cash_recover_[dt/pmdt/beel/lra]

Model Field calculated:

- lgd_cash_recovery_amt_[dt/pmdt/beel/lra]

Logic:

lgd_cash_recovery_amt_&type. = lgd_recovery_ead2 * lgd_cash_recover_&type.;
 &type. is replaced by [dt/beel/pmdt/lra] respectively.

Discount Factors

Model Fields required:

- lgd_disc_rate_cust

Parameters Fields required:

- lgd_disc_rate_monthly_[dt/pmdt/beel/lra]

- lgd_tts_[dt/pmdt/beel/lra]
- lgd_ttp_[dt/pmdt/beel/lra]

Model Field calculated:

- lgd_cust_disc_factor_sale_[dt/pmdt/beel/lra]
- lgd_cust_disc_factor_pip_[dt/pmdt/beel/lra]

Parameter Field calculated:

- lgd_reg_disc_factor_sale_[dt/pmdt/beel/lra]
- lgd_reg_disc_factor_pip_[dt/pmdt/beel/lra]

Logic:

/* Discount Factor - Sale to Default - Regulatory Discount Rate */

lgd_reg_disc_factor_sale_&type. = 1 / (lgd_disc_rate_monthly_&type.)*lgd_tts_&type.;

/* Discount Factor - Possession to Default - Regulatory Discount Rate */

lgd_reg_disc_factor_pip_&type. = 1 / (lgd_disc_rate_monthly_&type.)*lgd_ttp_&type.;

/* Discount Factor - Sale to Default - Customer Rate */

lgd_cust_disc_factor_sale_&type. = 1 / (lgd_disc_rate_cust)*lgd_tts_&type.;

/* Discount Factor - Possession to Default - Customer Rate */

lgd_cust_disc_factor_pip_&type. = 1 / (lgd_disc_rate_cust)*lgd_ttp_&type.;

&type. is replaced by [dt/beel/pmdt/lra] respectively.

Discounted Recoveries and Costs

Model Fields required:

- lgd_recovery_valuation_[dt/pmdt/beel/lra]
- lgd_cust_disc_factor_pip_[dt/pmdt/beel/lra]
- lgd_cust_disc_factor_sale_[dt/pmdt/beel/lra]
- lgd_cost_sale_variable_[dt/pmdt/beel/lra]
- lgd_cash_recovery_amt_[dt/pmdt/beel/lra]

Parameters Fields required:

- lgd_haircut_[dt/pmdt/beel/lra]
- lgd_shared_owner_adj_[dt/pmdt/beel/lra]
- lgd_reg_disc_factor_sale_[dt/pmdt/beel/lra]
- lgd_cost_list_[dt/pmdt/beel/lra]
- lgd_cost_poss_[dt/pmdt/beel/lra]
- lgd_cost_sale_[dt/pmdt/beel/lra]

Model Field calculated:

- lgd_sale_recov_disc_[dt/pmdt/beel/lra]
- lgd_cost_lit_disc_[dt/pmdt/beel/lra]
- lgd_cost_poss_sale_disc_[dt/pmdt/beel/lra]
- lgd_cash_recovery_amt_disc_[dt/pmdt/beel/lra]

Logic:

/* Apply discounting to sale recoveries - discounted at regulatory rate*/

```

lgd_sale_recov_disc_&type. = lgd_recovery_valuation_&type. * lgd_haircut_&type. *
lgd_shared_owner_adj_&type. * lgd_reg_disc_factor_sale_&type.;

/* Apply Discounting to Litigation Costs */

lgd_cost_lit_disc_&type. = lgd_cost_lit_&type. * lgd_cust_disc_factor_pip_&type.;

/* Apply Discounting to Possession and Sale Cost */

lgd_cost_poss_sale_disc_&type. = lgd_cost_poss_&type. * lgd_cust_disc_factor_pip_&type.
+ sum(lgd_cost_sale_&type., lgd_cost_sale_variable_&type.) *
lgd_cust_disc_factor_sale_&type.;

/* Apply Discounting to cash recovery */

lgd_cash_recovery_amt_disc_&type. = lgd_cash_recovery_amt_&type. *
lgd_reg_disc_factor_pip_&type.;

&type. is replaced by [dt/beel/pmdt/lra] respectively.

```

Sales Recoveries at Contractual Entitlement

Model Fields required:

- lgd_recovery_ead2 (derived in EAD model calculation) for dt, pmdt, lra
- lgd_recovery_ead1 (derived in EAD model calculation) for beel
- lgd_cash_recovery_amt_disc_[dt/pmdt/beel/lra]
- lgd_cost_lit_disc_[dt/pmdt/beel/lra]
- lgd_cost_poss_sale_disc_[dt/pmdt/beel/lra]
- lgd_sale_recov_disc_[dt/pmdt/beel/lra]

Model Field calculated:

- lgd_contractual_amt_[dt/pmdt/beel/lra]
- lgd_recovery_amt_disc_[dt/pmdt/beel/lra]

Logic:

```

/* Contractual Entitlement at Default */

lgd_contractual_amt_&type. = sum(&ead_type., -lgd_cash_recovery_amt_disc_&type.,
lgd_cost_lit_disc_&type., lgd_cost_poss_sale_disc_&type.);

/* Sale recoveries capped at contractual entitlement */

lgd_recovery_amt_disc_&type. = min(lgd_contractual_amt_&type.,
lgd_sale_recov_disc_&type.);

&type. is replaced by [dt/beel/pmdt/lra] respectively.

```

Severity at Possession - LGP

Model Fields required:

- lgd_recovery_ead2 (derived in EAD model calculation) for dt, pmdt, lra
- lgd_recovery_ead1 (derived in EAD model calculation) for beel
- lgd_cash_recovery_amt_disc_[dt/pmdt/beel/lra]
- lgd_recovery_amt_disc_[dt/pmdt/beel/lra]
- lgd_cost_lit_disc_[dt/pmdt/beel/lra]
- lgd_cost_poss_sale_disc_[dt/pmdt/beel/lra]

- lgd_cost_of_coll_[dt/pmdt/beel/lra]

Model Field calculated:

- lgd_severity_value_[dt/pmdt/beel/lra]
- lgd_severity_[dt/pmdt/beel/lra]
- lgd_ppgd_[dt/pmdt/beel/lra]
- lgd_sale_[dt/pmdt/beel/lra]

Logic:

```
lgd_severity_value_&type. = sum(&ead_type., -lgd_recovery_amt_disc_&type., -
lgd_cash_recovery_amt_disc_&type., lgd_cost_oss_sale_disc_&type.,
lgd_cost_lit_disc_&type., lgd_cost_of_coll_&type.);
```

```
/* LGP Floored at Zero */
```

```
if &ead_type. ne 0 then lgd_severity_&type. = max(0, lgd_severity_value_&type. /
&ead_type.);
```

```
else lgd_severity_&type. = .;
```

MoC for LGP:

```
%if %eval("&type"="DT" or "&type"="PMDT") %then %do;
    if default=0 then do;
        if lgd_recovery_segment='1' then
lgd_severity_&type=lgd_severity_&type*1.0021648044305346;
        if lgd_recovery_segment='2' then
lgd_severity_&type=lgd_severity_&type*1.0026263051756795;
        if lgd_recovery_segment='3' then
lgd_severity_&type=lgd_severity_&type*1.00088760560493018;
        if lgd_recovery_segment='4' then
lgd_severity_&type=lgd_severity_&type*1.00192037227076341;
        if lgd_recovery_segment='5' then
lgd_severity_&type=lgd_severity_&type*1.00107352398439396;
        if lgd_recovery_segment='6' then
lgd_severity_&type=lgd_severity_&type*1.00274353524016682;
    end;else do;
        if lgd_recovery_segment='7' then
lgd_severity_&type=lgd_severity_&type*1.00352277514490542;
        if lgd_recovery_segment='8' then
lgd_severity_&type=lgd_severity_&type*1.00341349485233911;
        if lgd_recovery_segment='9' then
lgd_severity_&type=lgd_severity_&type*1.00089131790443901;
        if lgd_recovery_segment='10' then
lgd_severity_&type=lgd_severity_&type*1.00286330484463146;
        if lgd_recovery_segment='11' then
lgd_severity_&type=lgd_severity_&type*1.00106952659810512;
```

```

                                if lgd_recovery_segment='12' then
lgd_severity_&type=lgd_severity_&type*1.00327176261431424;

                                if lgd_recovery_segment='13' then
lgd_severity_&type=lgd_severity_&type*1.0012843193;

                                end;

                                %end;%else %do;

/*                                LRA/BEEL                                */

                                if default=0 then do;

                                if lgd_recovery_segment='1' then
lgd_severity_&type=lgd_severity_&type*1.00085962534416137;

                                if lgd_recovery_segment='2' then
lgd_severity_&type=lgd_severity_&type*1.000553541333329156;

                                if lgd_recovery_segment='3' then
lgd_severity_&type=lgd_severity_&type*1.000373062975217872;

                                if lgd_recovery_segment='4' then
lgd_severity_&type=lgd_severity_&type*1.000524793627618789;

                                if lgd_recovery_segment='5' then
lgd_severity_&type=lgd_severity_&type*1.000470711682061679;

                                if lgd_recovery_segment='6' then
lgd_severity_&type=lgd_severity_&type*1.000988207345074194;

                                end;else do;

                                if lgd_recovery_segment='7' then
lgd_severity_&type=lgd_severity_&type*1.00123528596303534;

                                if lgd_recovery_segment='8' then
lgd_severity_&type=lgd_severity_&type*1.00077058322390833;

                                if lgd_recovery_segment='9' then
lgd_severity_&type=lgd_severity_&type*1.000355640087575302;

                                if lgd_recovery_segment='10' then
lgd_severity_&type=lgd_severity_&type*1.000738212763353399;

                                if lgd_recovery_segment='11' then
lgd_severity_&type=lgd_severity_&type*1.00043255018646472;

                                if lgd_recovery_segment='12' then
lgd_severity_&type=lgd_severity_&type*1.00107472710699345;

                                if lgd_recovery_segment='13' then
lgd_severity_&type=lgd_severity_&type*1.000502171;

                                end;

                                %end;

/* LGP * ppgd */

if lgd_severity_&type. = . then lgd_sale_&type. = .;

else lgd_sale_&type. = lgd_ppgd_&type. * lgd_severity_&type.;

```

&type. is replaced by [dt/beel/pmdt/lra] respectively.

LGD Cure

Model Fields required:

- lgd_recovery_ead2 (derived in EAD model calculation)

Parameter Fields required:

- lgd_lgc_[dt/pmdt/beel/lra]
- lgd_ppgd_[dt/pmdt/beel/lra]

Model Field calculated:

- lgd_cure_severity_value_[dt/pmdt/beel/lra]

Parameter Fields set:

- lgd_cure_severity_[dt/pmdt/beel/lra]
- lgd_cure_ppgd_[dt/pmdt/beel/lra]
- lgd_cure_[dt/pmdt/beel/lra]

Logic:

lgd_cure_severity_value_&type. = lgd_lgc_&type. * &ead_type.;

lgd_cure_severity_&type. = lgd_lgc_&type.;

lgd_cure_ppgd_&type. = max(0, sum(1, -lgd_ppgd_&type.));

lgd_cure_&type. = lgd_cure_ppgd_&type. * lgd_cure_severity_&type.;

&type. is replaced by [dt/beel/pmdt/lra] respectively.

Final LGDs

Drivers required:

- system_cd
- btl

Model Fields required:

- lgd_recovery_ead2 (derived in EAD model calculation)
- lgd_sale_[dt/pmdt/beel/lra]

Parameter Fields required:

- lgd_cure_[dt/pmdt/beel/lra]
- al_moc_[dt/pmdt/beel/lra]
- btl_moc_[dt/pmdt/beel/lra]

Model Field calculated:

- lgd_no_floor_al_[dt/pmdt/beel/lra]
- lgd_no_floor_[dt/pmdt/beel/lra]
- lgd_[dt/pmdt/beel/lra]

Logic:

	If a property has been sold but not yet written off and closed then do: <pre> /* 100% LGD if a property has been sold, but not yet written off and closed. */ if property_sale_date <= period_end_date and property_sale_date ^= . then do; lgd_&type. = 1; lgd_no_floor_&type.= 1; lgd_sale_&type. = 1; end; Otherwise: if system_cd = 'U' then lgd_no_floor_&type. = sum(lgd_sale_&type., lgd_cure_&type.) * al_moc_&type. * a9_moc_&type.; else lgd_no_floor_&type. = sum(lgd_sale_&type., lgd_cure_&type.) * a9_moc_&type.; /* Kept for reference */ lgd_&type. = max(0.05, lgd_no_floor_&type.); &type. is replaced by [dt/beel/pmdt/lra] respectively. /* NOW APPLIED AT GRADE LEVEL */ moc_c_factor = 1; if system_cd = 'U' then lgd_no_floor_no_moc_dt = lgd_no_floor_dt / al_moc_dt; else lgd_no_floor_no_moc_dt = lgd_no_floor_dt; lgd_no_floor_dt_mocc = moc_c_factor * lgd_no_floor_dt; lgd_dt = max(0.05, lgd_no_floor_dt_mocc); </pre>
--	---

3.3. EAD Functional Requirements

Scope of Mortgage Hybrid EAD Model

DDP Ref	DDP Mortgage Hybrid	System	Data Lake
Requirements	The filtering criteria to identify the range of application of the new Hybrid Mortgage EAD model is unchanged from the filters that are applied to the mortgage contracts table in the Data Lake to get the applicable contract population for calculating mortgages under the IRB model.		

Additional Information	The mortgage IRB perimeter is as follows: Portfolio: - cntr_portfolio = 'Mortgages' and cntr_subproduct_code in ('402','403','404','405','704','705','707','742','746','999') direct mapping from contract_mortgage_partitions (Retail).partenon_subproduct. cntr_portfolio of 'Mortgages' is set where the contract is sourced from common layer table retail_contracts_enriched.contract_mortgage_partitions, a filter is applied to contract_mortgage_partitions when loading main_contract_partitions where accounting_product_division is not 'STONEHAVEN_MORTGAGE' (also identified with Lender_code 61). Materiality (Balances > £100): - Where the contract is a mortgage i.e. main_contract_partitions.mortgage_cd is 'M' or 'F', then total of ('DRAWN_BALANCE'+ 'ACCRUED_INTEREST'+ 'UNDRAWN_BALANCE') from main_accounting_partitions is greater than 100. - Where the main_contract_partitions.mortgage_cd is 'N' i.e. a new offer the balance criteria is not applied. Asset Class: - asset_class not in (main_contract_partitions.cntr_asset_class = '2' or main_contract_partitions.cntr_asset_class='02') i.e. include where asset class is 1, 01 or null. IRB Account Status and Lender Code: - contract_mortgage_partitions.irb_account_status in ('OP', 'PO') or (contract_mortgage_partitions.irb_account_status = 'PC' and contract_mortgage_partitions.write_off_date is null) - AND NOT (source_system = 'ANMF' AND lender_code in (60,62)) 60 also = migrated UFSS Welcome contracts, 62 = migrated UFSS mortgage plus contracts and unsecured loan parts.
------------------------	---

Sourced EAD Model Risk Drivers

DDP Ref	DDP Mortgage Hybrid	System	Data Lake																																										
Requirements	Source and ensure inclusion of all EAD Model Risk Drivers in retail_capital_enriched.main_mortgage_driver_partitions table.																																												
Additional Information	<u>Introduction</u> This section provides a list of the EAD model risk drivers that already exist currently in the Data Lake (Either Avalon Capital Engine or Common Layer) and that are required in order to perform the EAD Model Calculation. <u>Sourced EAD Model Risk Drivers</u> All the fields are listed with their BDR or model development source name in the table below. The equivalent drivers in data lake should be used. This mapping should be checked by the implementation team. Data Lake tables referenced in table below are: Retail_capital_enriched.main_mortgage_driver_partitions (driver) Retail_capital_enriched.mortgage_ahc_output_partitions (ahc) retail_contracts_enriched.mortgage_loan_partitions.pricing (pricing) retail_contracts_enriched.contract_mortgage_partitions (contract) <table><tr><th>Dev/BDR Field</th><th>Existing Lake Field</th><th>Lake Table</th></tr><tr><td>holding_reference</td><td>holding_reference</td><td>driver</td></tr><tr><td>io_flag</td><td>io_flag</td><td>driver</td></tr><tr><td>maturity_months</td><td>approaching_maturity_mnth</td><td>driver</td></tr><tr><td>on_balance_sheet_exposure</td><td>on_balance_sheet_exposure</td><td>driver</td></tr><tr><td>offer_amt</td><td>offer_amt</td><td>driver</td></tr><tr><td>pra_status</td><td>status_pra</td><td>ahc</td></tr><tr><td>period_end_date</td><td>date_processed</td><td>driver</td></tr><tr><td>pra_default_entry_date</td><td>def_entry_date_pra</td><td>ahc</td></tr><tr><td>pra_arr_band</td><td>dpd_n_pra</td><td>driver</td></tr><tr><td>drawdown_avail_curr_amt</td><td>!</td><td>!</td></tr><tr><td>Interest_rate</td><td>Interest_rate</td><td>pricing</td></tr><tr><td>flexi_mortgage_ind</td><td>flexi_mortgage_ind</td><td>Driver</td></tr><tr><td>earliest_maturity</td><td>Maturity_date_npl</td><td>ahc</td></tr></table>			Dev/BDR Field	Existing Lake Field	Lake Table	holding_reference	holding_reference	driver	io_flag	io_flag	driver	maturity_months	approaching_maturity_mnth	driver	on_balance_sheet_exposure	on_balance_sheet_exposure	driver	offer_amt	offer_amt	driver	pra_status	status_pra	ahc	period_end_date	date_processed	driver	pra_default_entry_date	def_entry_date_pra	ahc	pra_arr_band	dpd_n_pra	driver	drawdown_avail_curr_amt	!	!	Interest_rate	Interest_rate	pricing	flexi_mortgage_ind	flexi_mortgage_ind	Driver	earliest_maturity	Maturity_date_npl	ahc
Dev/BDR Field	Existing Lake Field	Lake Table																																											
holding_reference	holding_reference	driver																																											
io_flag	io_flag	driver																																											
maturity_months	approaching_maturity_mnth	driver																																											
on_balance_sheet_exposure	on_balance_sheet_exposure	driver																																											
offer_amt	offer_amt	driver																																											
pra_status	status_pra	ahc																																											
period_end_date	date_processed	driver																																											
pra_default_entry_date	def_entry_date_pra	ahc																																											
pra_arr_band	dpd_n_pra	driver																																											
drawdown_avail_curr_amt	!	!																																											
Interest_rate	Interest_rate	pricing																																											
flexi_mortgage_ind	flexi_mortgage_ind	Driver																																											
earliest_maturity	Maturity_date_npl	ahc																																											

drawdown_limit_amt	!	!
mortgage_cd	mortgage_cd	Driver
Offset_balance	Offset_balance	Driver

Table 5: Mortgage Hybrid Sourced EAD Risk Drivers

Additional Notes:

drawdown_limit_amt and drawdown_avail_curr_amt

The mappings appear to be the same or very similar for these variables currently. This is something that needs to be resolved in implementation.

Derived EAD Model Risk Drivers

DDP Ref	DDP Mortgage Hybrid	System	Data Lake
Requirements	Calculate and add new EAD Model Risk Drivers to retail_capital_enriched.main_mortgage_driver_partitions table checking for any clashes with or duplication of existing variables.		

Additional Information

Introduction

This section provides a list of the EAD model risk drivers that are required to be newly calculated in the Data Lake (Either Avalon Capital Engine or Common Layer) and that are required in order to perform the EAD Model Calculation.

EAD Characteristics

The newly required fields are listed by name in the table below.

New Lake Field required as per Dev/BDR Field
drawdown_avail_curr_amt_adj
Max_int_rate
AM_IO
limit
usage
On_balance
interest_addon
int_rate_monthly
Int_rate_monthly_ec
Flexi
ROI
Ead_segment
Ccf_grade
Ccf_grade_label

Table 6: Mortgage Hybrid EAD Risk Drivers

This section provides details of where variables in the table above require some derivation rather than simply be sourced from the appropriate Data Lake table.

Additional Notes:

Max_int_rate

The max_int_rate must be obtained by transforming interest_rate from the pricing table. It is defined as the maximum interest rate applied to any portion of a loan. In SAS, this is as follows:

```
max(case when period_end_balance_amt > 0 and interest_rate > 0 then interest_rate else 0
end) as max_int_rate
group by contract id
```

drawdown avail curr amt adj

An adjusted drawdown available is calculated by removing the A&L offset balances as follows:

```
if substr(holding_reference, 1, 4) = 'UFSS' then do;
offset_balance = drawdown_avail_curr_amt;
```

	<pre> drawdown_avail_curr_amt_adj = 0; end; else do; drawdown_avail_curr_amt_adj = drawdown_avail_curr_amt; end; </pre> <u>On balance</u> The on-balance is first adjusted with the offset balance and then floored at 0. on_balance2 = on_balance_sheet_exposure + offset_balance; on_balance = max(0, on_balance2); <u>Limit</u> The limit is calculated as the sum of drawn and undrawn exposures associated with an account. limit = on_balance + drawdown_avail_curr_amt_adj; <u>Usage</u> The usage is calculated as the drawn portion of the limit. usage = on_balance / limit; <u>interest_addon</u> An addon is included to add to the monthly rate of maximum yearly interest to convert it to a downturn interest rate. In the code, this is defined as a macro variable. %let interest_addon = 1; <u>int_rate_monthly</u> The monthly rate of max_interest (yearly) is calculated, with an addon, currently set to 1% at the time of initial implementation as per variable interest_addon above, to convert the interest rate to a downturn interest rate. Int_rate_monthly = max(0, ((1 + ((max_int_rate + &interest_addon.) / 100)) ** (1/12)) - 1) <u>int_rate_monthly_ec</u> Same as int_rate_monthly, without the downturn addon. int_rate_monthly_ec = max(0, ((1 + ((max_int_rate) / 100)) ** (1/12)) - 1); <u>Flexi</u> The flexi segment is identified by the flexi mortgage indicator and the availability of drawdown. if flexi_mortgage_ind = '' then flexi_mortgage_ind = 'N'; if (flexi_mortgage_ind = 'Y') and (drawdown_avail_curr_amt_adj > 0) then flexi = 'Y'; else flexi = 'N'; <u>AM IO</u> Flexi approaching maturity, interest only accounts are identified and flagged as follows: if flexi = 'Y' then do; if (-9 <= intck('month', datepart(earliest_maturity), datepart(period_end_date)) <= 3) and (io_flag in ('I', 'P')) then AM_IO = 1; else AM_IO = 0;
--	---

Note: it is understood that AM accounts are identified separately for LGD/PD using the maturity_months field. The calculations should net the same results (not considering the IO part).

ROI

Flexi accounts in the ROI segment are identified by absolute and proportional limit uses.

if flexi = 'Y' then do;

if (drawdown_avail_curr_amt_adj / drawdown_limit_amt <= **0.025**) or (drawdown_avail_curr_amt_adj <= **500**) then do;

ROI = **1**;

else do;

ROI = **0**;

end;

EAD segment

Segment is chosen for the EAD model by the following logic. The segment should align to the EAD model calculation shown in the next section. The implementation code does things in a different order, where all of segmentation and EAD calculation is performed at the same time.

if pra_status not in ('PRO','PRR','DEF','PIP','SFS','WRO') and (mortgage_cd in ('M', 'F', 'N')) then do;

if (flexi = 'Y') and (mortgage_cd = 'M') then do;

if (ROI = **0**) and (AM_IO = **0**) then do;

ead_segment = 'FLEXI_CCF';

end;

else if AM_IO = **1** then do;

ead_segment = 'AM_IO';

end;

else if (ROI = **1**) and (AM_IO = **0**) then do;

ead_segment = 'ROI';

end;

else do;

ead_segment = 'ERROR';

end;

end;

```

else if mortgage_cd in ('N', 'F') then do;
    ead_segment = 'PIPELINE';
end;
else do;
    ead_segment = 'NON-FLEXI';
end;
end;

else if pra_status in ('PRO','PRR','DEF','PIP','SFS','WRO') then do;
    ead_segment = 'DEFAULT';
end;

```

CCF grade, CCF grade level

The CCF grades and labels for flexi accounts is assigned by the following. The usage split is defined in a macro variable and is not expected to change.

```

/*Usage segmentation split point*/
%let usage_split = 0.8671184124920526;

if (flexi = 'Y') and (mortgage_cd = 'M') then do;

    /*Select accounts not in RoI and not AM-IO and apply CCF values based on the final
    segmentation*/

    if (ROI = 0) and (AM_IO = 0) then do;

        ead_segment = 'FLEXI_CCF';

    /*Assign CCF grades*/

    /*Accounts with missing arrears or DPD90+ not flagged as defaults are assigned to DPD=0
    bands, which have the higher CCFs.*/

    Note – in the code, CCF grades and labels are assigned to other segments as ccf_grade=99
    and ccf_label="N/A" respectively. These assignments are not required for EAD calculation and
    are not required in implementations.

    if pra_arr_band in ('DPD030','DPD060','DPD090') then do;

    if usage >= &usage_split. then do;

```

```

ccf_grade = 1;

ccf_grade_label = "DPD>0 & Usage >= 0.867";

end;

else if usage < &usage_split. then do;

ccf_grade = 2;

ccf_grade_label = "DPD>0 & Usage < 0.867";

end;

end;

else if pra_arr_band not in ('DPD030','DPD060','DPD090') then do;

if usage >= &usage_split. then do;

ccf_grade = 3;

ccf_grade_label = "DPD=0 & Usage >= 0.867";

end;

else if usage < &usage_split. then do;

ccf_grade = 4;

ccf_grade_label = "DPD=0 & Usage < 0.867";

end;

end;

else do;

ccf_grade = 999;

ccf_grade_label = "MISSING";

end;

```

CCF_reg, CCF_reg MoC, CCF_ec

The regulatory pre and post MoC CCFs and the economic capital CCF are defined as follows. Macro variables for the CCFs are used but not expected to change.

/*DT CCFs with no AAs, No MoCs*/

```
%let ccf_reg_1 = 0.346406949423874;
```

```
%let ccf_reg_2 = 0.183020200828468;
```

```

%let ccf_reg_3 = 0.847377139179343;

%let ccf_reg_4 = 0.335791673907958;

/*CCFs including all AAs and MoCs*/

%let ccf_reg_moc_1 = 0.404637544661483;
%let ccf_reg_moc_2 = 0.268681796305314;
%let ccf_reg_moc_3 = 0.912675245337986;
%let ccf_reg_moc_4 = 0.383763837913987;

/*LRA CCFs without any AAs*/

%let ccf_lra_1 = 0.202595416;
%let ccf_lra_2 = 0.07635299;
%let ccf_lra_3 = 0.35451589;
%let ccf_lra_4 = 0.155201236;

/*AA for CCF economic capital*/

%let CCF_AA_LRA = 1 + 0.0004255288;

    if ccf_grade = 1 then do;
        ccf_reg = &ccf_reg_1.;
        ccf_reg_moc = &ccf_reg_moc_1.;
        ccf_ec = &ccf_lra_1. * (&CCF_AA_LRA.);
    end;

    else if ccf_grade = 2 then do;
        ccf_reg = &ccf_reg_2.;
        ccf_reg_moc = &ccf_reg_moc_2.;
        ccf_ec = &ccf_lra_2. * (&CCF_AA_LRA.);
    end;

```



```

else if ccf_grade = 3 then do;
    ccf_reg = &ccf_reg_3.;
    ccf_reg_moc = &ccf_reg_moc_3.;
    ccf_ec = &ccf_lra_3. * (&CCF_AA_LRA.);
end;

else if ccf_grade = 4 then do;
    ccf_reg = &ccf_reg_4.;
    ccf_reg_moc = &ccf_reg_moc_4.;
    ccf_ec = &ccf_lra_4. * (&CCF_AA_LRA.);
end;

```

Note – in the code, CCF values equal to 0 are assigned to other segments. These assignments are not required for EAD calculation and are not required in implementations.

EAD Model Calculation

DDP Ref	DDP Mortgage Hybrid	System	Data Lake
Requirements	Identification of segments and application of parameters/estimates for EAD model		

Additional Information	<u>Introduction</u> The calculation of the EAD Model pertains to the retail_capital_enriched.main_mortgage_model_partitions table in the Data Lake. A new version of this table is required to execute the Mortgage Hybrid table which will eventually replace the existing model table used to execute Mortgage Variable Scalar. The following variables will be required to be calculated for the Mortgage Hybrid version. These are: • EAD_est • Ead_est_final • Ead_est_ec • Lgd_recovery_ead1 • Lgd_recovery_ead2 All newly calculated fields mentioned below will pertain to a Hybrid version of the following Data Lake table: Retail_capital_enriched.main_mortgage_model_partitions (model) <u>EAD est, ead est final, ead est ec, ead reg</u> The EAD estimated for regulatory capital (pre-MoC), regulatory capital post-MoC, and economic capital are set below. The segmentation is aligned to the EAD_segment in the derived model risk drivers. A number of macro variables are used in the code, and their values are not expected to change. Drivers required: • Flexi • Roi • AM_IO • pra_arr_band • Mortgage_cf Parameters set: • Ead_est • Ead_est_final • Ead_est_ec • Ead_reg Logic: <pre> /*Roi DT AA + MoC value: taken from max_int_roi_v3.xlsx*/ %let roi_moc_value = 1 + 0.0030174485; /*ROI LRA AA value*/ </pre>
-------------------------------	---

```

%let roi_aa_value = 1 + 0.0002359967;

/*Non-Flexi DT AA + MoC value*/

%let non_flexi_moc_value = 1 + 0.0021795714;

/*Non-Flexi LRA AA value - unused when using current balance model*/

%let non_flexi_AA_value = 1 + 0.0021458048;

/*Arrears fee used in EAD calculation for RoI*/

%let arr_fee = 40;

/*DT pre-MoC, pre-AA Assumed drawdown for AM-IO accounts*/

%let assumed_dd = 1939.46;

/*post-AA, post-MoC assumed drawdown for AM-IO accounts, used for reg cap*/

%let assumed_dd_reg = 3601.06;

/*LRA, no AA applied, no MoC applied, used for econ cap*/

%let assumed_dd_ec = 630.12;

/*Stressed interest rate addon for RoI and Non-Flexi*/

%let interest_addon = 1;

/*Pipeline CCFs for further offers and new offers, ec and reg cap*/

/*Offers given 50% CF per article 166D/Article 111*/

/*Pipeline CCFs for further offers and new offers*/

%let pipe_FA_reg = 0.9091;

%let pipe_NO_reg = 0.9318;

/*Highest 12m post MoC - used for reg cap*/

%let pipe_FA_reg_moc = 0.9133;

%let pipe_NO_reg_moc = 0.9331;

/*LRA (no AAs or MoCs) - used for econ cap*/

%let pipe_FA_ec = 0.8730;

%let pipe_NO_ec = 0.8969;/*LRA (no AAs or MoCs) - used for econ cap*/

if (flexi = 'Y') and (mortgage_cd = 'M') then do;

  /*Select accounts not in RoI and not AM-IO and apply CCF values based on the final
  segmentation*/

```

```

if (ROI = 0) and (AM_IO = 0) then do;

/*Calculate EAD for CCF Flexis*/

ead_est = sum(on_balance,(ccf_reg * drawdown_avail_curr_amt_adj));

/*RC for flexis*/

ead_est_final = sum(on_balance,(ccf_reg_moc * drawdown_avail_curr_amt_adj));

/*EC for flexis */

ead_est_ec = sum(on_balance,(ccf_ec * drawdown_avail_curr_amt_adj));end;

else if AM_IO = 1 then do;

ead_est = sum(on_balance,&assumed_dd.);

ead_est_final = sum(on_balance,&assumed_dd_reg.);

ead_est_ec = sum(on_balance,&assumed_dd_ec.);

end;

else if (ROI = 1) and (AM_IO = 0) then do;

    ead_est =
(sum(on_balance_sheet_exposure,offset_balance,drawdown_avail_curr_amt_adj,(4 *
&arr_fee.))) * (sum(1,int_rate_monthly)**3);

    ead_est_final = (&roi_moc_value.) * ead_est;

    ead_est_EC = (&roi_aa_value.) *
(sum(on_balance_sheet_exposure,offset_balance,drawdown_avail_curr_amt_adj,(4 *
&arr_fee.))) * (sum(1,int_rate_monthly_ec)**3);

    end;

/*Pipeline*/

else if mortgage_cd in ('N', 'F') then do;

/*New offers*/

if mortgage_cd in ('N') then do;

    ead_est =
(sum((&pipe_NO_reg.*offer_amt),on_balance_sheet_exposure,offset_balance,drawdown_a
vail_curr_amt_adj,(4 * &arr_fee.))) * (sum(1,int_rate_monthly)**3);

/*B3.0*/

    ead_est_final =
(sum((&pipe_NO_reg_moc.*offer_amt),on_balance_sheet_exposure,offset_balance,drawdo
wn_avail_curr_amt_adj,(4 * &arr_fee.))) * (sum(1,int_rate_monthly_ec)**3);

```

```

    ead_est_ec =
sum((&pipe_NO_ec.*offer_amt),on_balance_sheet_exposure,offset_balance,drawdown_av
ail_curr_amt_adj);

    end;

    if mortgage_cd in ('F') then do;

        ead_est =
(sum((&pipe_FA_reg.*offer_amt),on_balance_sheet_exposure,offset_balance,drawdown_a
vail_curr_amt_adj,(4 * &arr_fee.))) * (sum(1,int_rate_monthly)**3);

        /*B3.0*/

        ead_est_final =
(sum((&pipe_FA_reg_moc.*offer_amt),on_balance_sheet_exposure,offset_balance,drawdo
wn_avail_curr_amt_adj,(4 * &arr_fee.))) * (sum(1,int_rate_monthly_ec)**3);

        ead_est_ec =
sum((&pipe_FA_ec.*offer_amt),on_balance_sheet_exposure,offset_balance,drawdown_ava
il_curr_amt_adj);

        end;

    end;

else do;

    /*Non-Flexi model*/

    ead_est =
(sum(on_balance_sheet_exposure,offset_balance,drawdown_avail_curr_amt_adj,(4 *
&arr_fee.))) * (sum(1,int_rate_monthly)**3);

    ead_est_final = (&non_flexi_moc_value.) * ead_est;

    ead_est_ec =
sum(on_balance_sheet_exposure,offset_balance,drawdown_avail_curr_amt_adj);

;end;

*set default EADs;

if pra_status in ('PRO','PRR','DEF','PIP','SFS','WRO') then do;
ead_est = on_balance_sheet_exposure;
ead_est_final = ead_est;
ead_est_ec = ead_est;
end;

ead_est=max(ead_est,on_balance_sheet_exposure,0);
ead_est_final=max(ead_est_final,on_balance_sheet_exposure,0);
ead_est_ec=max(ead_est_ec,on_balance_sheet_exposure,0);

```

	<pre> ead_reg = ead_est_final; <u>lgd_recovery_ead2, lgd_recovery_ead1</u> Calculated as the downturn EAD for all accounts. The only difference is for pipeline accounts, a 100% pipeline CCF is used. drivers required: • Ead_segment • On_balance_sheet_exposure • Offset_balance • Drawdown_avail_curr_amt_adj • Ead_est_final (from previous calculation) • Ead_est_ec /*LGD recovery EAD assumes completed offers*/ if ead_segment = 'PIPELINE' then do; lgd_recovery_ead2 = max(0,on_balance_sheet_exposure,(sum((1*offer_amt),on_balance_sheet_exposure,offset _balance,drawdown_avail_curr_amt_adj,(4 * &arr_fee.))) * (sum(1,int_rate_monthly)**3)); lgd_recovery_ead1 = max(0,on_balance_sheet_exposure,(sum(offer_amt,on_balance_sheet_exposure,offset_bal ance,drawdown_avail_curr_amt_adj))); else do; lgd_recovery_ead2 = ead_est_final; lgd_recovery_ead1 = ead_est_ec; end; </pre>

3.4. Calculation of LGD Floor

DDP Ref	DDP Mortgage Hybrid	System	Data Lake
Requirements	LGD Floor Calculation		

Additional Information	The following fields will be required in the calculation of the LGD Floor from the Data Lake: lgd_dt ead_reg The aim of the calculation is to adjust account level LGD_DTs so that at a portfolio level the LGD is a minimum 10%. It is an iterative process the stages of which are set out below. Some calculations are carried out at account level others at the portfolio level resulting in 1 value to be applied to all accounts, the level of the calculation is specified. Step 1 The Upper and Lower LGD bounds are set: LB=0.05 (Portfolio level) UP=0.1 (Portfolio level) Step 2 The portfolio level LGD weighted by EAD is calculated along with the total portfolio EAD: TOTAL_EAD = Sum(EAD_REG) (Portfolio level) PORT_LEVEL_LGD = Sum(LGD_DT*EAD_REG)/TOTAL_EAD (Portfolio level) If PORT_LEVEL_LGD is >10% then do: LGD_REG=LGD_DT; Else perform steps 3, 4 & 5 below: Step 3 Calculate CONSIDERED_FLOOR as the midpoint between the Upper and Lower bounds: CONSIDERED_FLOOR= (UB+LB)/2; (Portfolio Level) where LGD_DT is less than CONSIDERED_FLOOR then the LGD_DT_PORT_FLOOR is set to CONSIDERED_FLOOR: If LGD_DT<CONSIDERED_FLOOR THEN DO; LGD_DT_PORT_FLOOR=CONSIDERED_FLOOR; (Account level) PORT_FLOOR_FLAG=1; (Account level) END; ELSE DO; LGD_DT_PORT_FLOOR=LGD_DT; (Account level) PORT_FLOOR_FLAG=0; (Account level) END; Re-calculate the portfolio level LGD weighted by EAD: EAD_WTD_DT_LGD=(LGD_DT_PORT_FLOOR*EAD_REG)/TOTAL_EAD; (Account level) PORT_LEVEL_LGD=SUM(EAD_WTD_DT_LGD) (Portfolio level)
-------------------------------	--

	If $0.0999 \leq \text{PORT_LEVEL_LGD} \leq 0.1001$ Then stop iterating else follow step 4 below: Step 4 Where $\text{PORT_LEVEL_LGD} < 0.0999$: Repeat Step 3 replacing the lower bound (LB) with the latest CONSIDERED_FLOOR value Or Where $\text{PORT_LEVEL_LGD} > 0.1001$: Repeat Step 3 replacing the upper bound (UB) with the latest CONSIDERED_FLOOR value Keep iterating through Step 4 until $0.0999 \leq \text{PORT_LEVEL_LGD} \leq 0.1001$ NOTE: when UB and LB are updated this new value should be retained for subsequent iterations i.e. they do not revert back to 0.05 and 0.1. Step 5 If $\text{PORT_LEVEL_LGD} < 0.1$ and $\text{PORT_FLOOR_FLAG} = 1$ then $\text{LGD_DT_PORT_FLOOR} = \text{CEIL}(\text{LGD_DT_PORT_FLOOR} * 1000) / 1000$; (Account level) $\text{LGD_REG} = \text{MAX}(\text{LGD_DT}, \text{CONSIDERED_FLOOR})$; (Account level) LGD_REG should then be added to the Hybrid version of the following Data Lake table: Retail_capital_enriched.main_mortgage_model_partitions (model)

3.5. Affected Channels

<i>Internet Individuals</i>	N/A as only Capital Calculation affected.
<i>Internet Companies</i>	N/A as only Capital Calculation affected.
<i>Offices</i>	N/A as only Capital Calculation affected.
<i>Call Centre</i>	N/A as only Capital Calculation affected.
<i>Extranet</i>	N/A as only Capital Calculation affected.

*Self Banking*

N/A as only Capital Calculation affected.

Mobile Banking

N/A as only Capital Calculation affected.

Intranet

N/A as only Capital Calculation affected.

Others (Identify)

Capital Calculations and Reporting.

3.6. Security Requirements

As per existing Avalon Capital Calculation Engine.

4. Distribution List for Review and Sign Off

Name	Department	Sign Off	Review

5. Appendix
